

Wavelet systems in database

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1 Wavelet system: Q0_IN_02_ST_DF_NS_FL

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.238186$

$a : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$

Wavelet Masks: $VM(b_1) = 0$,

$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$,

$\tilde{b}_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{-1} \\ 1 \end{pmatrix}$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Haar
Dual Refinable Mask Info	Delta Function
Symmetry Info	No Symmetry
Short Code	Q0_IN_02_ST_DF_NS_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

2 Wavelet system: Q0_IN_02_ST_BI_D2SX_OB

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.238186$

$$a : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.238186$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	ONB
Refinable Mask Info	Haar
Dual Refinable Mask Info	Haar
Symmetry Info	D2 With Center (1/2,0)
Short Code	Q0_IN_02_ST_BI_D2SX_OB

Source: Haar wavelet

3 Wavelet system: Q0_IN_03_ST_DF_NS_FL

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.778195$

$$a : \quad \frac{1}{32} \begin{pmatrix} \boxed{16} & 7 \\ 9 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} 0 & \boxed{-9} \\ -7 & 16 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Standard
Dual Refinable Mask Info	Delta Function
Symmetry Info	No Symmetry
Short Code	Q0_IN_03_ST_DF_NS_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

4 Wavelet system: Q0_IN_03_ST_1C_NS_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.778195$

$$a : \quad \frac{1}{32} \begin{pmatrix} \boxed{16} & 7 \\ 9 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.238186$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & \boxed{-9} \\ -7 & 23 \\ -7 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Standard
Dual Refinable Mask Info	1-Compatible In 2-point Standard
Symmetry Info	No Symmetry
Short Code	Q0_IN_03_ST_1C_NS_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

5 Wavelet system: Q0_IN_03_ST_BI_NS_FS

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.778195$

$$a : \quad \frac{1}{32} \begin{pmatrix} \boxed{16} & 7 \\ 9 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = -0.126148$

$$\tilde{a} : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & -9 \\ 0 & \boxed{48} & 16 \\ -7 & 16 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$,

$$b_1 : \quad \frac{1}{64} \begin{pmatrix} 0 & \boxed{-16} & -7 \\ -16 & 48 & 0 \\ -9 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} 0 & \boxed{-9} \\ -7 & 16 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Dual Wavelet Frame In Sobolev Spaces
Refinable Mask Info	Best In 3-point Standard
Dual Refinable Mask Info	Biorthogonal In Cube-1
Symmetry Info	No Symmetry
Short Code	Q0_IN_03_ST_BI_NS_FS

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

6 Wavelet system: Q0_IN_03_ST_BI_NS_FR

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.778195$

$$a : \quad \frac{1}{32} \begin{pmatrix} \boxed{16} & 7 \\ 9 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.0458584$

$$\tilde{a} : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & -135 \\ 0 & \boxed{694} & 240 \\ -175 & 400 & 72 \\ 0 & 56 & -128 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$,

$$b_1 : \quad \frac{1}{1024} \begin{pmatrix} 128 & 56 & 0 \\ 72 & \boxed{-400} & -175 \\ -240 & 694 & 0 \\ -135 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} 0 & \boxed{-9} \\ -7 & 16 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Best In 3-point Standard
Dual Refinable Mask Info	Biorthogonal In Special Support
Symmetry Info	No Symmetry
Short Code	Q0_IN_03_ST_BI_NS_FR

Source: Created by the algorithm in paper "Symmetric multivariate wavelets" by Karakazyan S., Skopina M., Tchobanou M. in Int. J. Wavelets Multiresolut. Inf. Process. 2009 7 (3) P. 313-340.

7 Wavelet system: Q0_IN_03_HL_DF_D2_FL

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.09619$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	Delta Function
Symmetry Info	D2
Short Code	Q0_IN_03_HL_DF_D2_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

8 Wavelet system: Q0_IN_03_HL_1C_NS_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.09619$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.238186$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 3 \\ -1 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	1-Compatible In 2-point Standard
Symmetry Info	No Symmetry
Short Code	Q0_IN_03_HL_1C_NS_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

9 Wavelet system: Q0_IN_03_HL_2C_D2_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.09619$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.09619$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 \\ \boxed{2} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} -1 \\ \boxed{-2} \\ 6 \\ -2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	2-Compatible In 3-point Line
Symmetry Info	D2
Short Code	Q0_IN_03_HL_2C_D2_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

10 Wavelet system: Q0_IN_03_HL_BI_D2_FS

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.09619$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = -0.458953$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} -1 \\ 2 \\ \boxed{6} \\ 2 \\ -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$,

$$b_1 : \quad \frac{1}{8} \begin{pmatrix} -1 \\ \boxed{-2} \\ 6 \\ -2 \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Dual Wavelet Frame In Sobolev Spaces
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	Biorthogonal In 5-point Line
Symmetry Info	D2
Short Code	Q0_IN_03_HL_BI_D2_FS

Source: Created by the algorithm in paper "Symmetric multivariate wavelets" by Karakazyan S., Skopina M., Tchobanov M. in Int. J. Wavelets Multiresolut. Inf. Process. 2009 7 (3) P. 313-340.

11 Wavelet system: Q0_IN_05_R1_DF_D4_FL

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.57764$

$$a : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} 0 & \boxed{-1} & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-1
Dual Refinable Mask Info	Delta Function
Symmetry Info	D4
Short Code	Q0_IN_05_R1_DF_D4_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

12 Wavelet system: Q0_IN_05_R1_2C_D2_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.57764$

$$a : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.09619$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 \\ \boxed{2} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{16} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{-2} & -1 \\ -2 & 14 & -2 \\ -1 & -2 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-1
Dual Refinable Mask Info	2-Compatible In 3-point Line
Symmetry Info	D2
Short Code	Q0_IN_05_R1_2C_D2_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

13 Wavelet system: Q0_IN_05_R1_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.57764$

$$a : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{4} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -2 & \boxed{-4} & -2 & 0 \\ -1 & -4 & 28 & -4 & -1 \\ 0 & -2 & -4 & -2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q0_IN_05_R1_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

14 Wavelet system: Q0_IN_05_R1_BI_D4_FR

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.57764$

$$a : \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4$, $sm_2(\tilde{a}) = 0.575457$

$$\tilde{a} : \frac{1}{512} \begin{pmatrix} 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 8 & 3 & 0 & 0 & 0 \\ 0 & 0 & 10 & -20 & -16 & -20 & 10 & 0 & 0 \\ 0 & 3 & -20 & -38 & 96 & -38 & -20 & 3 & 0 \\ -2 & 8 & -16 & 96 & \boxed{416} & 96 & -16 & 8 & -2 \\ 0 & 3 & -20 & -38 & 96 & -38 & -20 & 3 & 0 \\ 0 & 0 & 10 & -20 & -16 & -20 & 10 & 0 & 0 \\ 0 & 0 & 0 & 3 & 8 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 4$,

$$b_1 : \frac{1}{512} \begin{pmatrix} 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & -8 & 3 & 0 & 0 & 0 \\ 0 & 0 & 10 & 20 & -16 & 20 & 10 & 0 & 0 \\ 0 & 3 & 20 & -38 & \boxed{-96} & -38 & 20 & 3 & 0 \\ -2 & -8 & -16 & -96 & 416 & -96 & -16 & -8 & -2 \\ 0 & 3 & 20 & -38 & -96 & -38 & 20 & 3 & 0 \\ 0 & 0 & 10 & 20 & -16 & 20 & 10 & 0 & 0 \\ 0 & 0 & 0 & 3 & -8 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$,

$$\tilde{b}_1 : \frac{1}{8} \begin{pmatrix} 0 & \boxed{-1} & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Best In Rhomb-1
Dual Refinable Mask Info	Biorthogonal In Rhomb-4
Symmetry Info	D4
Short Code	Q0_IN_05_R1_BI_D4_FR

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

15 Wavelet system: Q0_IN_13_C2_DF_D4_FL

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.44792$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & -1 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & 10 & \boxed{32} & 10 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \left(\boxed{1} \right)$$

Wavelet Masks: $VM(b_1) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{64} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & \boxed{-10} & 0 & 1 \\ 0 & -10 & 32 & -10 & 0 \\ 1 & 0 & -10 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	Delta Function
Symmetry Info	D4
Short Code	Q0_IN_13_C2_DF_D4_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

16 Wavelet system: Q0_IN_13_C2_2C_D2_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.44792$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & -1 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & 10 & \boxed{32} & 10 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.09619$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 \\ \boxed{2} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{128} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 2 & -10 & 2 & 1 \\ 2 & -9 & \boxed{-20} & -9 & 2 \\ 2 & -20 & 108 & -20 & 2 \\ 2 & -9 & -20 & -9 & 2 \\ 1 & 2 & -10 & 2 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	2-Compatible In 3-point Line
Symmetry Info	D2
Short Code	Q0_IN_13_C2_2C_D2_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

17 Wavelet system: Q0_IN_13_C2_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.44792$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & -1 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & 10 & \boxed{32} & 10 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{4} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & -8 & 4 & 2 & 0 \\ 1 & 4 & -18 & \boxed{-40} & -18 & 4 & 1 \\ 0 & -8 & -40 & 216 & -40 & -8 & 0 \\ 1 & 4 & -18 & -40 & -18 & 4 & 1 \\ 0 & 2 & 4 & -8 & 4 & 2 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q0_IN_13_C2_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

18 Wavelet system: Q0_IN_13_C2_4C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.44792$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & -1 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & 10 & \boxed{32} & 10 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.235724$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} -1 & 2 & -1 \\ 2 & \boxed{12} & 2 \\ -1 & 2 & -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{16} \begin{pmatrix} 1 & -2 & 1 \\ -2 & \boxed{4} & -2 \\ 1 & -2 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{512} \begin{pmatrix} 0 & -1 & 2 & -2 & 2 & -1 & 0 \\ -1 & 4 & 21 & -16 & 21 & 4 & -1 \\ 2 & 21 & -36 & \boxed{-102} & -36 & 21 & 2 \\ -2 & -16 & -102 & 432 & -102 & -16 & -2 \\ 2 & 21 & -36 & -102 & -36 & 21 & 2 \\ -1 & 4 & 21 & -16 & 21 & 4 & -1 \\ 0 & -1 & 2 & -2 & 2 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	4-Compatible In Cube-1
Symmetry Info	D4
Short Code	Q0_IN_13_C2_4C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

19 Wavelet system: Q0_IN_13_C2_BI_D4_FS

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.44792$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & -1 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & 10 & \boxed{32} & 10 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = -0.185251$

$$\tilde{a} : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & -8 & 0 & 2 & 0 \\ 1 & 0 & -18 & 32 & -18 & 0 & 1 \\ 0 & -8 & 32 & \boxed{216} & 32 & -8 & 0 \\ 1 & 0 & -18 & 32 & -18 & 0 & 1 \\ 0 & 2 & 0 & -8 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$,

$$b_1 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & -8 & 0 & 2 & 0 \\ 1 & 0 & -18 & -32 & -18 & 0 & 1 \\ 0 & -8 & \boxed{-32} & 216 & -32 & -8 & 0 \\ 1 & 0 & -18 & -32 & -18 & 0 & 1 \\ 0 & 2 & 0 & -8 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{64} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & -10 & 0 & 1 \\ 0 & \boxed{-10} & 32 & -10 & 0 \\ 1 & 0 & -10 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Dual Wavelet Frame In Sobolev Spaces
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	Biorthogonal In Cube-3
Symmetry Info	D4
Short Code	Q0_IN_13_C2_BI_D4_FS

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

20 Wavelet system: Q0_IN_17_R3S_4C_D4_FR

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4, sm_2(a) = 2.35143$

$$a : \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ -1 & 0 & 39 & \boxed{128} & 39 & 0 & -1 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4, sm_2(\tilde{a}) = 2.35143$

$$\tilde{a} : \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ -1 & 0 & 39 & \boxed{128} & 39 & 0 & -1 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2, VM(b_2) = 2, VM(b_3) = 4,$

$$b_1 : \frac{1}{32} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -2 & \boxed{0} & -2 & 0 \\ -1 & 0 & 12 & 0 & -1 \\ 0 & -2 & 0 & -2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

$$b_2 : \frac{1}{2048} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & -64 & 0 & 6 & 0 & 0 \\ 0 & 4 & 0 & -136 & \boxed{0} & -136 & 0 & 4 & 0 \\ 1 & 0 & -64 & 0 & 740 & 0 & -64 & 0 & 1 \\ 0 & 4 & 0 & -136 & 0 & -136 & 0 & 4 & 0 \\ 0 & 0 & 6 & 0 & -64 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$b_3 : \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & \boxed{0} & 3 & 0 & 0 \\ 0 & 3 & 0 & -39 & 0 & 3 & 0 \\ 1 & 0 & -39 & 128 & -39 & 0 & 1 \\ 0 & 3 & 0 & -39 & 0 & 3 & 0 \\ 0 & 0 & 3 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2, VM(\tilde{b}_3) = 4,$

$$\tilde{b}_1 : \frac{1}{2048} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & -64 & 0 & 6 & 0 & 0 \\ 0 & 4 & 0 & -136 & \boxed{0} & -136 & 0 & 4 & 0 \\ 1 & 0 & -64 & 0 & 740 & 0 & -64 & 0 & 1 \\ 0 & 4 & 0 & -136 & 0 & -136 & 0 & 4 & 0 \\ 0 & 0 & 6 & 0 & -64 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \frac{1}{32} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -2 & \boxed{0} & -2 & 0 \\ -1 & 0 & 12 & 0 & -1 \\ 0 & -2 & 0 & -2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_3 : \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & \boxed{0} & 3 & 0 & 0 \\ 0 & 3 & 0 & -39 & 0 & 3 & 0 \\ 1 & 0 & -39 & 128 & -39 & 0 & 1 \\ 0 & 3 & 0 & -39 & 0 & 3 & 0 \\ 0 & 0 & 3 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Custom
Dual Refinable Mask Info	4-Compatible In Rhomb-3
Symmetry Info	D4
Short Code	Q0_IN_17_R3S_4C_D4_FR

Source: Created by the algorithm in paper "Compactly Supported Multivariate Pairs of Dual Wavelet Frames Obtained by Convolution" by M. Ehler, 2008, IJWMIP 6 (2), 183-208.

21 Wavelet system: Q0_IN_17_R3S_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.35143$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ -1 & 0 & 39 & \boxed{128} & 39 & 0 & -1 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{4} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 4 & 4 & 0 & 0 & 0 \\ 0 & 0 & 6 & 12 & -32 & 12 & 6 & 0 & 0 \\ 0 & 4 & 12 & -72 & \boxed{-156} & -72 & 12 & 4 & 0 \\ 1 & 4 & -32 & -156 & 868 & -156 & -32 & 4 & 1 \\ 0 & 4 & 12 & -72 & -156 & -72 & 12 & 4 & 0 \\ 0 & 0 & 6 & 12 & -32 & 12 & 6 & 0 & 0 \\ 0 & 0 & 0 & 4 & 4 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom in Rhomb-3
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q0_IN_17_R3S_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

22 Wavelet system: Q0_NI_09_C1_2C_D2_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.09619$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & -1 \\ -2 & -4 & -2 \\ -2 & \boxed{24} & -2 \\ -2 & -4 & -2 \\ -1 & 0 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{16} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{-2} & -1 \\ -2 & 14 & -2 \\ -1 & -2 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In 3-point Line
Symmetry Info	D2
Short Code	Q0_NI_09_C1_2C_D2_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

23 Wavelet system: Q0_NI_09_C1_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & -1 & 0 \\ -1 & -4 & -6 & -4 & -1 \\ 0 & -6 & \boxed{48} & -6 & 0 \\ -1 & -4 & -6 & -4 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -2 & \boxed{-4} & -2 & 0 \\ -1 & -4 & 28 & -4 & -1 \\ 0 & -2 & -4 & -2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q0_NI_09_C1_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

24 Wavelet system: Q0_NI_09_C1_4C_D4_FL

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = -1.45625$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} -3 & 2 & -3 \\ 2 & \boxed{20} & 2 \\ -3 & 2 & -3 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{128} \begin{pmatrix} 3 & -2 & 6 & -2 & 3 \\ -2 & -8 & -12 & -8 & -2 \\ 6 & -12 & \boxed{60} & -12 & 6 \\ -2 & -8 & -12 & -8 & -2 \\ 3 & -2 & 6 & -2 & 3 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & 3 & -2 & 3 & 0 \\ 3 & -4 & \boxed{-14} & -4 & 3 \\ -2 & -14 & 56 & -14 & -2 \\ 3 & -4 & -14 & -4 & 3 \\ 0 & 3 & -2 & 3 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	4-Compatible In Cube-1
Symmetry Info	D4
Short Code	Q0_NI_09_C1_4C_D4_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

25 Wavelet system: Q0_NI_09_C1_4C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.686754$

$$\tilde{a} : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & -9 & 0 & 0 \\ 0 & 6 & 8 & 6 & 0 \\ -9 & 8 & \boxed{44} & 8 & -9 \\ 0 & 6 & 8 & 6 & 0 \\ 0 & 0 & -9 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 9 & 0 & 9 & 0 & 0 \\ 0 & -6 & -8 & 24 & -8 & -6 & 0 \\ 9 & -8 & -50 & -48 & -50 & -8 & 9 \\ 0 & 24 & -48 & \boxed{312} & -48 & 24 & 0 \\ 9 & -8 & -50 & -48 & -50 & -8 & 9 \\ 0 & -6 & -8 & 24 & -8 & -6 & 0 \\ 0 & 0 & 9 & 0 & 9 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 3 & -8 & 3 & 0 & 0 \\ 0 & 3 & -16 & \boxed{-47} & -16 & 3 & 0 \\ 9 & -8 & -47 & 224 & -47 & -8 & 9 \\ 0 & 3 & -16 & -47 & -16 & 3 & 0 \\ 0 & 0 & 3 & -8 & 3 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	4-Compatible In Rhomb-2
Symmetry Info	D4
Short Code	Q0_NI_09_C1_4C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

26 Wavelet system: Q0_NI_04_SS_2C_D1SX_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 1.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 0 & \boxed{1} & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.238186$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 1$,

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} 0 & 0 & -1 \\ 0 & \boxed{2} & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} 0 & 0 & -1 \\ 0 & \boxed{4} & -1 \\ 1 & -4 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Special
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D1 With Center (1/2,0)
Short Code	Q0_NI_04_SS_2C_D1SX_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

27 Wavelet system: Q0_NI_06_C1_2C_D2SX_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 3, sm_2(a) = 3.0$

$$a : \quad \frac{1}{8} \begin{pmatrix} 1 & \boxed{2} & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 0.238186$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 3,$

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} -1 & 0 & -1 \\ -1 & \boxed{4} & -1 \\ -1 & 4 & -1 \\ -1 & 0 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} -1 & 0 & -1 \\ -1 & \boxed{8} & -1 \\ 1 & -8 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D2 With Center (1/2,0)
Short Code	Q0_NI_06_C1_2C_D2SX_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

28 Wavelet system: Q0_NI_06_C1_2C_D2SX_FLL2_2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 3$, $sm_2(a) = 3.0$

$$a : \quad \frac{1}{8} \begin{pmatrix} 1 & \boxed{2} & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 3$, $sm_2(\tilde{a}) = 3.0$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 1 & \boxed{2} & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 1$,

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 3$,

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} -1 & -2 & -2 & -2 & -1 \\ -1 & -6 & \boxed{22} & -6 & -1 \\ -1 & -6 & 22 & -6 & -1 \\ -1 & -2 & -2 & -2 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} -1 & -2 & -2 & -2 & -1 \\ -1 & -2 & \boxed{30} & -2 & -1 \\ 1 & 2 & -30 & 2 & 1 \\ 1 & 2 & 2 & 2 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D2 With Center (1/2,0)
Short Code	Q0_NI_06_C1_2C_D2SX_FLL2_2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

29 Wavelet system: Q0_NI_06_C1_3C_D2SX_FL

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 3, sm_2(a) = 3.0$

$$a : \quad \frac{1}{8} \begin{pmatrix} 1 & \boxed{2} & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = -1.91008$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{7} & -1 \\ -1 & 7 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4, VM(\tilde{b}_2) = 3,$

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & -7 & 6 & -7 & 1 \\ 1 & -2 & \boxed{6} & -2 & 1 \\ 1 & -2 & 6 & -2 & 1 \\ 1 & -7 & 6 & -7 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & -7 & 2 & -7 & 1 \\ 1 & -8 & \boxed{34} & -8 & 1 \\ -1 & 8 & -34 & 8 & -1 \\ -1 & 7 & -2 & 7 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	3-Compatible In Rhomb-2
Symmetry Info	D2 With Center (1/2,0)
Short Code	Q0_NI_06_C1_3C_D2SX_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

30 Wavelet system: Q0_NI_04_C1_2C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} -1 & -1 & 0 \\ -1 & \boxed{6} & -1 \\ 0 & -1 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} 0 & \boxed{-1} & -1 \\ -1 & 6 & -1 \\ -1 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q0_NI_04_C1_2C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

31 Wavelet system: Q0_NI_04_C1_2C_D4SXY_TF

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2, VM(b_2) = 1, VM(b_3) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} 1 & \boxed{-1} \\ -1 & 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ -1 & -1 \end{pmatrix}$$

$$b_3 : \quad \frac{1}{4} \begin{pmatrix} -1 & \boxed{1} \\ -1 & 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 1, VM(\tilde{b}_3) = 1,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} 1 & \boxed{-1} \\ -1 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ -1 & -1 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{4} \begin{pmatrix} -1 & \boxed{1} \\ -1 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Tight Wavelet Frame
Refinable Mask Info	Best In Rhomb-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q0_NI_04_C1_2C_D4SXY_TF

Source: Created by the algorithm in paper "Symmetric Canonical Quincunx Tight Framelets with High Vanishing Moments and Smoothness" by B. Han, Q. Jiang, Z. Shen, X. Zhuang, 2015, Mathematics of Computation, 87 (309), 347-379

32 Wavelet system: Q0_NI_12_R2_2C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & \boxed{6} & 6 & 1 \\ 1 & 6 & 6 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & -1 & -1 & 0 \\ 0 & -6 & -7 & -2 & -1 \\ -1 & -7 & \boxed{52} & -7 & -1 \\ -1 & -2 & -7 & -6 & 0 \\ 0 & -1 & -1 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & -1 & 0 & 0 \\ -1 & -2 & \boxed{-7} & -6 & 0 \\ -1 & -7 & 52 & -7 & -1 \\ 0 & -6 & -7 & -2 & -1 \\ 0 & 0 & -1 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-2
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q0_NI_12_R2_2C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

33 Wavelet system: Q0_NI_12_R2_4C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & \boxed{6} & 6 & 1 \\ 1 & 6 & 6 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.103406$

$$\tilde{a} : \quad \frac{1}{32} \begin{pmatrix} 0 & -3 & -3 & 0 \\ -3 & \boxed{14} & 14 & -3 \\ -3 & 14 & 14 & -3 \\ 0 & -3 & -3 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 21 & 4 & -11 & 6 & 0 \\ 0 & 21 & -78 & -77 & 8 & -11 & 3 \\ 3 & 4 & -77 & \boxed{356} & -77 & 4 & 3 \\ 3 & -11 & 8 & -77 & -78 & 21 & 0 \\ 0 & 6 & -11 & 4 & 21 & 0 & 0 \\ 0 & 0 & 3 & 3 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 3 & 3 & 0 & 0 & 0 \\ 0 & 6 & -11 & 4 & 21 & 0 & 0 \\ 3 & -11 & 8 & \boxed{-77} & -78 & 21 & 0 \\ 3 & 4 & -77 & 356 & -77 & 4 & 3 \\ 0 & 21 & -78 & -77 & 8 & -11 & 3 \\ 0 & 0 & 21 & 4 & -11 & 6 & 0 \\ 0 & 0 & 0 & 3 & 3 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-2
Dual Refinable Mask Info	4-Compatible In Rhomb-2
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q0_NI_12_R2_4C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

34 Wavelet system: Q0_NI_16_C2_2C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 6$, $sm_2(a) = 6.0$

$$a : \quad \frac{1}{64} \begin{pmatrix} 1 & 3 & 3 & 1 \\ 3 & \boxed{9} & 9 & 3 \\ 3 & 9 & 9 & 3 \\ 1 & 3 & 3 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{128} \begin{pmatrix} -1 & -1 & -3 & -3 & 0 \\ -1 & -10 & -12 & -6 & -3 \\ -3 & -12 & \boxed{110} & -12 & -3 \\ -3 & -6 & -12 & -10 & -1 \\ 0 & -3 & -3 & -1 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{128} \begin{pmatrix} 0 & -3 & -3 & -1 & -1 \\ -3 & -6 & \boxed{-12} & -10 & -1 \\ -3 & -12 & 110 & -12 & -3 \\ -1 & -10 & -12 & -6 & -3 \\ -1 & -1 & -3 & -3 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q0_NI_16_C2_2C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

35 Wavelet system: Q0_NI_16_C2_4C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 6$, $sm_2(a) = 6.0$

$$a : \quad \frac{1}{64} \begin{pmatrix} 1 & 3 & 3 & 1 \\ 3 & \boxed{9} & 9 & 3 \\ 3 & 9 & 9 & 3 \\ 1 & 3 & 3 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4$, $sm_2(\tilde{a}) = 1.3524$

$$\tilde{a} : \quad \frac{1}{64} \begin{pmatrix} -5 & -3 & -3 & -5 \\ -3 & \boxed{27} & 27 & -3 \\ -3 & 27 & 27 & -3 \\ -5 & -3 & -3 & -5 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{2048} \begin{pmatrix} 5 & 3 & 18 & 14 & 9 & 15 & 0 \\ 3 & 18 & 9 & -36 & -27 & 18 & 15 \\ 18 & 9 & -207 & -270 & -108 & -27 & 9 \\ 14 & -36 & -270 & \boxed{1608} & -270 & -36 & 14 \\ 9 & -27 & -108 & -270 & -207 & 9 & 18 \\ 15 & 18 & -27 & -36 & 9 & 18 & 3 \\ 0 & 15 & 9 & 14 & 18 & 3 & 5 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{2048} \begin{pmatrix} 0 & 15 & 9 & 14 & 18 & 3 & 5 \\ 15 & 18 & -27 & -36 & 9 & 18 & 3 \\ 9 & -27 & -108 & \boxed{-270} & -207 & 9 & 18 \\ 14 & -36 & -270 & 1608 & -270 & -36 & 14 \\ 18 & 9 & -207 & -270 & -108 & -27 & 9 \\ 3 & 18 & 9 & -36 & -27 & 18 & 15 \\ 5 & 3 & 18 & 14 & 9 & 15 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	4-Compatible In Cube-2
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q0_NI_16_C2_4C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

36 Wavelet system: Q0_NI_08_SS_4C_D4SXY_TF

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.03654$

$$a : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4$, $sm_2(\tilde{a}) = 3.03654$

$$\tilde{a} : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 4$, $VM(b_2) = 2$, $VM(b_3) = 2$,

$$b_1 : \quad \begin{pmatrix} -0.031 & 0 & 0 & 0.031 \\ 0 & 0.281 & \boxed{-0.281} & 0 \\ 0 & -0.281 & 0.281 & 0 \\ 0.031 & 0 & 0 & -0.031 \end{pmatrix}$$

$$b_2 : \quad \begin{pmatrix} -0.008 & 0 & 0 & -0.008 \\ 0 & \boxed{0.133} & 0.133 & 0 \\ 0 & -0.242 & -0.242 & 0 \\ 0.117 & 0 & 0 & 0.117 \end{pmatrix}$$

$$b_3 : \quad \begin{pmatrix} 0.117 & 0 & 0 & -0.117 \\ 0 & -0.242 & \boxed{0.242} & 0 \\ 0 & -0.133 & 0.133 & 0 \\ 0.008 & 0 & 0 & -0.008 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 2$, $VM(\tilde{b}_3) = 2$,

$$\tilde{b}_1 : \quad \begin{pmatrix} -0.031 & 0 & 0 & 0.031 \\ 0 & 0.281 & \boxed{-0.281} & 0 \\ 0 & -0.281 & 0.281 & 0 \\ 0.031 & 0 & 0 & -0.031 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \begin{pmatrix} -0.008 & 0 & 0 & -0.008 \\ 0 & \boxed{0.133} & 0.133 & 0 \\ 0 & -0.242 & -0.242 & 0 \\ 0.117 & 0 & 0 & 0.117 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \begin{pmatrix} 0.117 & 0 & 0 & -0.117 \\ 0 & -0.242 & \boxed{0.242} & 0 \\ 0 & -0.133 & 0.133 & 0 \\ 0.008 & 0 & 0 & -0.008 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Tight Wavelet Frame
Refinable Mask Info	Custom In Cube-2
Dual Refinable Mask Info	4-Compatible In Cube-2
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q0_NI_08_SS_4C_D4SXY_TF

Source: Created by the algorithm in paper "Symmetric Canonical Quincunx Tight Framelets with High Vanishing Moments and Smoothness" by B. Han, Q. Jiang, Z. Shen, X. Zhuang, 2015, Mathematics of Computation, 87 (309), 347-379

37 Wavelet system: Q0_NI_08_SS_2C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.03654$

$$a : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{64} \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & -8 & -9 & 0 & 0 \\ 0 & -9 & \boxed{46} & -9 & 0 \\ 0 & 0 & -9 & -8 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & \boxed{-9} & -8 & 1 \\ 0 & -9 & 46 & -9 & 0 \\ 1 & -8 & -9 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom In Cube-2
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q0_NI_08_SS_2C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

38 Wavelet system: Q0_NI_08_SS_4C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.03654$

$$a : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4$, $sm_2(\tilde{a}) = 3.03654$

$$\tilde{a} : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} -1 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 18 & 9 & 0 & 9 & 0 & 0 \\ 0 & 9 & -63 & -81 & 0 & 9 & 0 \\ -1 & 0 & -81 & \boxed{348} & -81 & 0 & -1 \\ 0 & 9 & 0 & -81 & -63 & 9 & 0 \\ 0 & 0 & 9 & 0 & 9 & 18 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & -1 \\ 0 & 0 & 9 & 0 & 9 & 18 & 0 \\ 0 & 9 & 0 & \boxed{-81} & -63 & 9 & 0 \\ -1 & 0 & -81 & 348 & -81 & 0 & -1 \\ 0 & 9 & -63 & -81 & 0 & 9 & 0 \\ 0 & 18 & 9 & 0 & 9 & 0 & 0 \\ -1 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom In Cube-2
Dual Refinable Mask Info	4-Compatible In Cube-2
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q0_NI_08_SS_4C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

39 Wavelet system: Q0_NI_09_SS_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.42789$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 1 & 4 & \boxed{12} & 4 & 1 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{128} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -4 & -1 & 0 & 0 \\ 0 & -1 & 0 & -13 & 0 & -1 & 0 \\ -1 & -4 & -13 & \boxed{80} & -13 & -4 & -1 \\ 0 & -1 & 0 & -13 & 0 & -1 & 0 \\ 0 & 0 & -1 & -4 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -2 & \boxed{-4} & -2 & 0 \\ -1 & -4 & 28 & -4 & -1 \\ 0 & -2 & -4 & -2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Special
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q0_NI_09_SS_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

40 Wavelet system: Q0_NI_09_SS_4C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.42789$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 1 & 4 & \boxed{12} & 4 & 1 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.686754$

$$\tilde{a} : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & -9 & 0 & 0 \\ 0 & 6 & 8 & 6 & 0 \\ -9 & 8 & \boxed{44} & 8 & -9 \\ 0 & 6 & 8 & 6 & 0 \\ 0 & 0 & -9 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -6 & -8 & -6 & 0 & 0 & 0 \\ 0 & 0 & 18 & -8 & 64 & -8 & 18 & 0 & 0 \\ 0 & -6 & -8 & -84 & -104 & -84 & -8 & -6 & 0 \\ 9 & -8 & 64 & -104 & \boxed{532} & -104 & 64 & -8 & 9 \\ 0 & -6 & -8 & -84 & -104 & -84 & -8 & -6 & 0 \\ 0 & 0 & 18 & -8 & 64 & -8 & 18 & 0 & 0 \\ 0 & 0 & 0 & -6 & -8 & -6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 3 & -8 & 3 & 0 & 0 \\ 0 & 3 & -16 & \boxed{-47} & -16 & 3 & 0 \\ 9 & -8 & -47 & 224 & -47 & -8 & 9 \\ 0 & 3 & -16 & -47 & -16 & 3 & 0 \\ 0 & 0 & 3 & -8 & 3 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Special
Dual Refinable Mask Info	4-Compatible In Rhomb-2
Symmetry Info	D4
Short Code	Q0_NI_09_SS_4C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

41 Wavelet system: Q0_NI_13_R2S_4C_D4_FR

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.91803$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 8 & 2 & 0 \\ 1 & 8 & \boxed{20} & 8 & 1 \\ 0 & 2 & 8 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57796$

$$\tilde{a} : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 0 & -9 & 0 & 0 & 0 \\ 0 & 0 & -27 & 12 & -27 & 0 & 0 \\ 0 & -27 & 24 & 127 & 24 & -27 & 0 \\ -9 & 12 & 127 & \boxed{112} & 127 & 12 & -9 \\ 0 & -27 & 24 & 127 & 24 & -27 & 0 \\ 0 & 0 & -27 & 12 & -27 & 0 & 0 \\ 0 & 0 & 0 & -9 & 0 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$, $VM(b_2) = 2$,

$$b_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & -5 & 0 & 0 \\ 0 & -10 & -40 & -10 & 0 \\ -5 & -40 & \boxed{220} & -40 & -5 \\ 0 & -10 & -40 & -10 & 0 \\ 0 & 0 & -5 & 0 & 0 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -3 & -8 & -3 & 0 & 0 \\ 0 & -3 & -16 & \boxed{7} & -16 & -3 & 0 \\ -1 & -8 & 7 & 96 & 7 & -8 & -1 \\ 0 & -3 & -16 & 7 & -16 & -3 & 0 \\ 0 & 0 & -3 & -8 & -3 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{320} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 9 & -4 & 9 & 0 & 0 \\ 0 & 9 & -8 & -85 & -8 & 9 & 0 \\ 3 & -4 & -85 & \boxed{304} & -85 & -4 & 3 \\ 0 & 9 & -8 & -85 & -8 & 9 & 0 \\ 0 & 0 & 9 & -4 & 9 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & -3 & 0 & 0 \\ 0 & -6 & \boxed{4} & -6 & 0 \\ -3 & 4 & 20 & 4 & -3 \\ 0 & -6 & 4 & -6 & 0 \\ 0 & 0 & -3 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Custom in Rhomb-2
Dual Refinable Mask Info	4-Compatible In Rhomb-3
Symmetry Info	D4
Short Code	Q0_NI_13_R2S_4C_D4_FR

Source: Created by the algorithm in paper "Bi-frames with 4-fold axial symmetry for quadrilateral surface multiresolution processing" by Q. Jiang, 2010, CAM 234 (12), 3303-3325

42 Wavelet system: Q0_NI_13_R2S_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.91803$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 8 & 2 & 0 \\ 1 & 8 & \boxed{20} & 8 & 1 \\ 0 & 2 & 8 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4$, $sm_2(\tilde{a}) = 4.0$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & -1 & -2 & -1 & 0 & 0 \\ 0 & -2 & -6 & -8 & -6 & -2 & 0 \\ -1 & -6 & -30 & -50 & -30 & -6 & -1 \\ -2 & -8 & -50 & \boxed{424} & -50 & -8 & -2 \\ -1 & -6 & -30 & -50 & -30 & -6 & -1 \\ 0 & -2 & -6 & -8 & -6 & -2 & 0 \\ 0 & 0 & -1 & -2 & -1 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & -2 & -1 & 0 \\ -1 & -4 & \boxed{-6} & -4 & -1 \\ -2 & -6 & 56 & -6 & -2 \\ -1 & -4 & -6 & -4 & -1 \\ 0 & -1 & -2 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom in Rhomb-2
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D4
Short Code	Q0_NI_13_R2S_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

43 Wavelet system: Q0_NI_13_R2S_4C_D4_FL

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.91803$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 8 & 2 & 0 \\ 1 & 8 & \boxed{20} & 8 & 1 \\ 0 & 2 & 8 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = -1.45625$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} -3 & 2 & -3 \\ 2 & \boxed{20} & 2 \\ -3 & 2 & -3 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 3 & -2 & 3 & 0 & 0 \\ 0 & 6 & -6 & -8 & -6 & 6 & 0 \\ 3 & -6 & 26 & -50 & 26 & -6 & 3 \\ -2 & -8 & -50 & \boxed{136} & -50 & -8 & -2 \\ 3 & -6 & 26 & -50 & 26 & -6 & 3 \\ 0 & 6 & -6 & -8 & -6 & 6 & 0 \\ 0 & 0 & 3 & -2 & 3 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & 3 & -2 & 3 & 0 \\ 3 & -4 & \boxed{-14} & -4 & 3 \\ -2 & -14 & 56 & -14 & -2 \\ 3 & -4 & -14 & -4 & 3 \\ 0 & 3 & -2 & 3 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom in Rhomb-2
Dual Refinable Mask Info	4-Compatible In Cube-1
Symmetry Info	D4
Short Code	Q0_NI13_R2S_4C_D4_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

44 Wavelet system: Q1_IN_02_ST_DF_NS_FL

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 1.0$

a : $\frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$\tilde{a}$: $\begin{pmatrix} 1 \end{pmatrix}$

Wavelet Masks: $VM(b_1) = 0$,

b_1 : $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$,

$\tilde{b}_1$: $\frac{1}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Haar
Dual Refinable Mask Info	Delta Function
Symmetry Info	No Symmetry
Short Code	Q1_IN_02_ST_DF_NS_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

45 Wavelet system: Q1_IN_02_ST_BI_D2SX_OB

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 1.0$

$$a : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	ONB
Refinable Mask Info	Haar
Dual Refinable Mask Info	Haar
Symmetry Info	D2 With Center (1/2,0)
Short Code	Q1_IN_02_ST_BI_D2SX_OB

Source: Haar wavelet

46 Wavelet system: Q1_IN_03_HL_DF_D2_FL

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	Delta Function
Symmetry Info	D2
Short Code	Q1_IN_03_HL_DF_D2_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

47 Wavelet system: Q1_IN_03_HL_1C_NS_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1, VM(\tilde{b}_2) = 1,$

$$\tilde{b}_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 3 \\ -1 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	1-Compatible In 2-point Standard
Symmetry Info	No Symmetry
Short Code	Q1_IN_03_HL_1C_NS_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

48 Wavelet system: Q0_IN_03_HL_2C_D2_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 \\ \boxed{2} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} -1 \\ \boxed{-2} \\ 6 \\ -2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	2-Compatible In 3-point Line
Symmetry Info	D2
Short Code	Q0_IN_03_HL_2C_D2_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

49 Wavelet system: Q1_IN_03_HL_BI_D2_FR

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.440765$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} -1 \\ 2 \\ \boxed{6} \\ 2 \\ -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$,

$$b_1 : \quad \frac{1}{8} \begin{pmatrix} -1 \\ \boxed{-2} \\ 6 \\ -2 \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	Biorthogonal In 5-point Line
Symmetry Info	D2
Short Code	Q1_IN_03_HL_BI_D2_FR

Source: Created by the algorithm in paper "Symmetric multivariate wavelets" by Karakazyan S., Skopina M., Tchobanou M. in Int. J. Wavelets Multiresolut. Inf. Process. 2009 7 (3) P. 313-340.

50 Wavelet system: Q1_IN_13_C2_DF_D1_FL

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.6677$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 0 & -2 & 0 \\ -2 & 0 & 5 & 0 & 1 \\ 0 & 5 & \boxed{16} & 5 & 0 \\ 1 & 0 & 5 & 0 & -2 \\ 0 & -2 & 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \left(\boxed{1} \right)$$

Wavelet Masks: $VM(b_1) = 0$,

$$b_1 : \quad \left(\begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix} \right)$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} 0 & -1 & 0 & 2 & 0 \\ 2 & 0 & \boxed{-5} & 0 & -1 \\ 0 & -5 & 16 & -5 & 0 \\ -1 & 0 & -5 & 0 & 2 \\ 0 & 2 & 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	Delta Function
Symmetry Info	D1
Short Code	Q1_IN_13_C2_DF_D1_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

51 Wavelet system: Q1_IN_13_C2_2C_D1_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.6677$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 0 & -2 & 0 \\ -2 & 0 & 5 & 0 & 1 \\ 0 & 5 & \boxed{16} & 5 & 0 \\ 1 & 0 & 5 & 0 & -2 \\ 0 & -2 & 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 \\ \boxed{2} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & 2 & 0 \\ 2 & -2 & -5 & 4 & -1 \\ 4 & -6 & \boxed{-10} & -3 & -2 \\ 1 & -10 & 54 & -10 & 1 \\ -2 & -3 & -10 & -6 & 4 \\ -1 & 4 & -5 & -2 & 2 \\ 0 & 2 & 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D1
Short Code	Q1_IN_13_C2_2C_D1_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

52 Wavelet system: Q1_IN_13_C2_4C_D1_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.6677$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 0 & -2 & 0 \\ -2 & 0 & 5 & 0 & 1 \\ 0 & 5 & \boxed{16} & 5 & 0 \\ 1 & 0 & 5 & 0 & -2 \\ 0 & -2 & 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.235724$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} -1 & 2 & -1 \\ 2 & \boxed{12} & 2 \\ -1 & 2 & -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{16} \begin{pmatrix} 1 & -2 & 1 \\ -2 & \boxed{4} & -2 \\ 1 & -2 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{256} \begin{pmatrix} 0 & 1 & -2 & -1 & 4 & -2 & 0 \\ -2 & 2 & -9 & -8 & 30 & 2 & 1 \\ 4 & 30 & -18 & \boxed{-51} & -18 & -9 & -2 \\ -1 & -8 & -51 & 216 & -51 & -8 & -1 \\ -2 & -9 & -18 & -51 & -18 & 30 & 4 \\ 1 & 2 & 30 & -8 & -9 & 2 & -2 \\ 0 & -2 & 4 & -1 & -2 & 1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	4-Compatible In Cube-1
Symmetry Info	D1
Short Code	Q1_IN_13_C2_4C_D1_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

53 Wavelet system: Q1_IN_13_C2_4C_D1_FLL2_2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.6677$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 0 & -2 & 0 \\ -2 & 0 & 5 & 0 & 1 \\ 0 & 5 & \boxed{16} & 5 & 0 \\ 1 & 0 & 5 & 0 & -2 \\ 0 & -2 & 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.83392$

$$\tilde{a} : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & -17 & 0 & 0 \\ 0 & 18 & 32 & 18 & 0 \\ -17 & 32 & \boxed{124} & 32 & -17 \\ 0 & 18 & 32 & 18 & 0 \\ 0 & 0 & -17 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 17 & 0 & 0 \\ 0 & -18 & -32 & -18 & 0 \\ 17 & -32 & \boxed{132} & -32 & 17 \\ 0 & -18 & -32 & -18 & 0 \\ 0 & 0 & 17 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4096} \begin{pmatrix} 0 & 0 & 0 & 17 & 0 & -34 & 0 & 0 & 0 \\ 0 & 0 & -52 & -32 & 103 & 64 & 53 & 0 & 0 \\ 0 & 53 & 32 & -127 & -128 & 242 & 32 & -52 & 0 \\ -34 & 64 & 242 & -288 & \boxed{-714} & -288 & -127 & -32 & 17 \\ 0 & 103 & -128 & -714 & 3456 & -714 & -128 & 103 & 0 \\ 17 & -32 & -127 & -288 & -714 & -288 & 242 & 64 & -34 \\ 0 & -52 & 32 & 242 & -128 & -127 & 32 & 53 & 0 \\ 0 & 0 & 53 & 64 & 103 & -32 & -52 & 0 & 0 \\ 0 & 0 & 0 & -34 & 0 & 17 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	4-Compatible In Rhomb-2
Symmetry Info	D1
Short Code	Q1_IN_13_C2_4C_D1_FLL2_2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

54 Wavelet system: Q1_IN_13_C2_BI_D1_FR

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.6677$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 0 & -2 & 0 \\ -2 & 0 & 5 & 0 & 1 \\ 0 & 5 & \boxed{16} & 5 & 0 \\ 1 & 0 & 5 & 0 & -2 \\ 0 & -2 & 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.0628943$

$$\tilde{a} : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & 1 & 0 & 2 & 0 \\ 2 & 0 & -3 & 0 & -6 & 0 & -1 \\ 0 & -5 & 16 & \boxed{54} & 16 & -5 & 0 \\ -1 & 0 & -6 & 0 & -3 & 0 & 2 \\ 0 & 2 & 0 & 1 & 0 & -1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$,

$$b_1 : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & 1 & 0 & 2 & 0 \\ 2 & 0 & -3 & \boxed{0} & -6 & 0 & -1 \\ 0 & -5 & -16 & 54 & -16 & -5 & 0 \\ -1 & 0 & -6 & 0 & -3 & 0 & 2 \\ 0 & 2 & 0 & 1 & 0 & -1 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} 0 & -1 & 0 & 2 & 0 \\ 2 & 0 & \boxed{-5} & 0 & -1 \\ 0 & -5 & 16 & -5 & 0 \\ -1 & 0 & -5 & 0 & 2 \\ 0 & 2 & 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	Biorthogonal In Cube-3
Symmetry Info	D1
Short Code	Q1_IN_13_C2_BI_D1_FR

Source: Created by the algorithm in paper "Symmetric multivariate wavelets" by Karakazyan S., Skopina M., Tchobanov M. in Int. J. Wavelets Multiresolut. Inf. Process. 2009 7 (3) P. 313-340.

55 Wavelet system: Q1_IN_17_R3S_4C_D4_FR

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4, sm_2(a) = 2.35143$

$$a : \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ -1 & 0 & 39 & \boxed{128} & 39 & 0 & -1 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4, sm_2(\tilde{a}) = 2.35143$

$$\tilde{a} : \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ -1 & 0 & 39 & \boxed{128} & 39 & 0 & -1 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2, VM(b_2) = 2, VM(b_3) = 4,$

$$b_1 : \frac{1}{32} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -2 & \boxed{0} & -2 & 0 \\ -1 & 0 & 12 & 0 & -1 \\ 0 & -2 & 0 & -2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

$$b_2 : \frac{1}{2048} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & -64 & 0 & 6 & 0 & 0 \\ 0 & 4 & 0 & -136 & \boxed{0} & -136 & 0 & 4 & 0 \\ 1 & 0 & -64 & 0 & 740 & 0 & -64 & 0 & 1 \\ 0 & 4 & 0 & -136 & 0 & -136 & 0 & 4 & 0 \\ 0 & 0 & 6 & 0 & -64 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$b_3 : \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & \boxed{0} & 3 & 0 & 0 \\ 0 & 3 & 0 & -39 & 0 & 3 & 0 \\ 1 & 0 & -39 & 128 & -39 & 0 & 1 \\ 0 & 3 & 0 & -39 & 0 & 3 & 0 \\ 0 & 0 & 3 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2, VM(\tilde{b}_3) = 4,$

$$\tilde{b}_1 : \frac{1}{2048} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & -64 & 0 & 6 & 0 & 0 \\ 0 & 4 & 0 & -136 & \boxed{0} & -136 & 0 & 4 & 0 \\ 1 & 0 & -64 & 0 & 740 & 0 & -64 & 0 & 1 \\ 0 & 4 & 0 & -136 & 0 & -136 & 0 & 4 & 0 \\ 0 & 0 & 6 & 0 & -64 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \frac{1}{32} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -2 & \boxed{0} & -2 & 0 \\ -1 & 0 & 12 & 0 & -1 \\ 0 & -2 & 0 & -2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_3 : \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & \boxed{0} & 3 & 0 & 0 \\ 0 & 3 & 0 & -39 & 0 & 3 & 0 \\ 1 & 0 & -39 & 128 & -39 & 0 & 1 \\ 0 & 3 & 0 & -39 & 0 & 3 & 0 \\ 0 & 0 & 3 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Custom in Rhomb-3
Dual Refinable Mask Info	4-Compatible In Rhomb-3
Symmetry Info	D4
Short Code	Q1_IN_17_R3S_4C_D4_FR

Source: Created by the algorithm in paper "Compactly Supported Multivariate Pairs of Dual Wavelet Frames Obtained by Convolution" by M. Ehler, 2008, IJWMIP 6 (2), 183-208.

56 Wavelet system: Q1_IN_17_R3S_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.35143$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ -1 & 0 & 39 & \boxed{128} & 39 & 0 & -1 \\ 0 & -3 & 0 & 39 & 0 & -3 & 0 \\ 0 & 0 & -3 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{4} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 4 & 4 & 0 & 0 & 0 \\ 0 & 0 & 6 & 12 & -32 & 12 & 6 & 0 & 0 \\ 0 & 4 & 12 & -72 & \boxed{-156} & -72 & 12 & 4 & 0 \\ 1 & 4 & -32 & -156 & 868 & -156 & -32 & 4 & 1 \\ 0 & 4 & 12 & -72 & -156 & -72 & 12 & 4 & 0 \\ 0 & 0 & 6 & 12 & -32 & 12 & 6 & 0 & 0 \\ 0 & 0 & 0 & 4 & 4 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom in Rhomb-3
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q1_IN_17_R3S_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

57 Wavelet system: Q1_NI_09_C1_2C_D2_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.09619$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & -1 \\ -2 & -4 & -2 \\ -2 & \boxed{24} & -2 \\ -2 & -4 & -2 \\ -1 & 0 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{16} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{-2} & -1 \\ -2 & 14 & -2 \\ -1 & -2 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In 3-point Line
Symmetry Info	D2
Short Code	Q1_NI_09_C1_2C_D2_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

58 Wavelet system: Q1_NI_09_C1_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & 0 & -1 & 0 \\ -1 & -4 & -6 & -4 & -1 \\ 0 & -6 & \boxed{48} & -6 & 0 \\ -1 & -4 & -6 & -4 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -2 & \boxed{-4} & -2 & 0 \\ -1 & -4 & 28 & -4 & -1 \\ 0 & -2 & -4 & -2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q1_NI_09_C1_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

59 Wavelet system: Q1_NI_09_C1_4C_D4_FL

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = -1.45625$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} -3 & 2 & -3 \\ 2 & \boxed{20} & 2 \\ -3 & 2 & -3 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{128} \begin{pmatrix} 3 & -2 & 6 & -2 & 3 \\ -2 & -8 & -12 & -8 & -2 \\ 6 & -12 & \boxed{60} & -12 & 6 \\ -2 & -8 & -12 & -8 & -2 \\ 3 & -2 & 6 & -2 & 3 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & 3 & -2 & 3 & 0 \\ 3 & -4 & \boxed{-14} & -4 & 3 \\ -2 & -14 & 56 & -14 & -2 \\ 3 & -4 & -14 & -4 & 3 \\ 0 & 3 & -2 & 3 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	4-Compatible In Cube-1
Symmetry Info	D4
Short Code	Q1_NI_09_C1_4C_D4_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

60 Wavelet system: Q1_NI_09_C1_4C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.686754$

$$\tilde{a} : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & -9 & 0 & 0 \\ 0 & 6 & 8 & 6 & 0 \\ -9 & 8 & \boxed{44} & 8 & -9 \\ 0 & 6 & 8 & 6 & 0 \\ 0 & 0 & -9 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 9 & 0 & 9 & 0 & 0 \\ 0 & -6 & -8 & 24 & -8 & -6 & 0 \\ 9 & -8 & -50 & -48 & -50 & -8 & 9 \\ 0 & 24 & -48 & \boxed{312} & -48 & 24 & 0 \\ 9 & -8 & -50 & -48 & -50 & -8 & 9 \\ 0 & -6 & -8 & 24 & -8 & -6 & 0 \\ 0 & 0 & 9 & 0 & 9 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 3 & -8 & 3 & 0 & 0 \\ 0 & 3 & -16 & \boxed{-47} & -16 & 3 & 0 \\ 9 & -8 & -47 & 224 & -47 & -8 & 9 \\ 0 & 3 & -16 & -47 & -16 & 3 & 0 \\ 0 & 0 & 3 & -8 & 3 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	4-Compatible In Rhomb-2
Symmetry Info	D4
Short Code	Q1_NI_09_C1_4C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

61 Wavelet system: Q1_NI_04_SS_2C_D1SX_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 1.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 & \boxed{1} & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.5$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 1$,

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 & 0 & 0 \\ -1 & \boxed{2} & 0 \\ 0 & 2 & -1 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} -1 & 0 & 0 \\ -1 & \boxed{4} & 0 \\ 0 & -4 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Special
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D1 With Center (1/2,0)
Short Code	Q1_NI_04_SS_2C_D1SX_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

62 Wavelet system: Q1_NI_06_C1_2C_D2SX_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 3, sm_2(a) = 3.0$

$$a : \quad \frac{1}{8} \begin{pmatrix} 1 & \boxed{2} & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 0.5$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 3,$

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} -1 & 0 & -1 \\ -1 & \boxed{4} & -1 \\ -1 & 4 & -1 \\ -1 & 0 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} -1 & 0 & -1 \\ -1 & \boxed{8} & -1 \\ 1 & -8 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D2 With Center (1/2,0)
Short Code	Q1_NI_06_C1_2C_D2SX_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

63 Wavelet system: Q1_NI_06_C1_2C_D2SX_FLL2_2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 3$, $sm_2(a) = 3.0$

$$a : \quad \frac{1}{8} \begin{pmatrix} 1 & \boxed{2} & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 3$, $sm_2(\tilde{a}) = 3.0$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 1 & \boxed{2} & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 1$,

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 3$,

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} -1 & -2 & -2 & -2 & -1 \\ -1 & -6 & \boxed{22} & -6 & -1 \\ -1 & -6 & 22 & -6 & -1 \\ -1 & -2 & -2 & -2 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} -1 & -2 & -2 & -2 & -1 \\ -1 & -2 & \boxed{30} & -2 & -1 \\ 1 & 2 & -30 & 2 & 1 \\ 1 & 2 & 2 & 2 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D2 With Center (1/2,0)
Short Code	Q1_NI_06_C1_2C_D2SX_FLL2_2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

64 Wavelet system: Q1_NI_06_C1_3C_D2SX_FL

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 3, sm_2(a) = 3.0$

$$a : \quad \frac{1}{8} \begin{pmatrix} 1 & \boxed{2} & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = -1.27917$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{7} & -1 \\ -1 & 7 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4, VM(\tilde{b}_2) = 3,$

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & -7 & 6 & -7 & 1 \\ 1 & -2 & \boxed{6} & -2 & 1 \\ 1 & -2 & 6 & -2 & 1 \\ 1 & -7 & 6 & -7 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & -7 & 2 & -7 & 1 \\ 1 & -8 & \boxed{34} & -8 & 1 \\ -1 & 8 & -34 & 8 & -1 \\ -1 & 7 & -2 & 7 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	3-Compatible In Rhomb-2
Symmetry Info	D2 With Center (1/2,0)
Short Code	Q1_NI_06_C1_3C_D2SX_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

65 Wavelet system: Q1_NI_04_C1_2C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} -1 & -1 & 0 \\ -1 & \boxed{6} & -1 \\ 0 & -1 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} 0 & \boxed{-1} & -1 \\ -1 & 6 & -1 \\ -1 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q1_NI_04_C1_2C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

66 Wavelet system: Q1_NI_04_C1_2C_D4SXY_TF

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2, VM(b_2) = 1, VM(b_3) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} 1 & \boxed{-1} \\ -1 & 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ -1 & -1 \end{pmatrix}$$

$$b_3 : \quad \frac{1}{4} \begin{pmatrix} -1 & \boxed{1} \\ -1 & 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 1, VM(\tilde{b}_3) = 1,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} 1 & \boxed{-1} \\ -1 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ -1 & -1 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{4} \begin{pmatrix} -1 & \boxed{1} \\ -1 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Tight Wavelet Frame
Refinable Mask Info	Best In Rhomb-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q1_NI_04_C1_2C_D4SXY_TF

Source: Created by the algorithm in paper "Symmetric Canonical Quincunx Tight Framelets with High Vanishing Moments and Smoothness" by B. Han, Q. Jiang, Z. Shen, X. Zhuang, 2015, Mathematics of Computation, 87 (309), 347-379

67 Wavelet system: Q1_NI_12_R2_2C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & \boxed{6} & 6 & 1 \\ 1 & 6 & 6 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & -1 & -1 & 0 \\ 0 & -6 & -7 & -2 & -1 \\ -1 & -7 & \boxed{52} & -7 & -1 \\ -1 & -2 & -7 & -6 & 0 \\ 0 & -1 & -1 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & -1 & 0 & 0 \\ -1 & -2 & \boxed{-7} & -6 & 0 \\ -1 & -7 & 52 & -7 & -1 \\ 0 & -6 & -7 & -2 & -1 \\ 0 & 0 & -1 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-2
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q1_NI_12_R2_2C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

68 Wavelet system: Q1_NI_12_R2_4C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & \boxed{6} & 6 & 1 \\ 1 & 6 & 6 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.103406$

$$\tilde{a} : \quad \frac{1}{32} \begin{pmatrix} 0 & -3 & -3 & 0 \\ -3 & \boxed{14} & 14 & -3 \\ -3 & 14 & 14 & -3 \\ 0 & -3 & -3 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 21 & 4 & -11 & 6 & 0 \\ 0 & 21 & -78 & -77 & 8 & -11 & 3 \\ 3 & 4 & -77 & \boxed{356} & -77 & 4 & 3 \\ 3 & -11 & 8 & -77 & -78 & 21 & 0 \\ 0 & 6 & -11 & 4 & 21 & 0 & 0 \\ 0 & 0 & 3 & 3 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 3 & 3 & 0 & 0 & 0 \\ 0 & 6 & -11 & 4 & 21 & 0 & 0 \\ 3 & -11 & 8 & \boxed{-77} & -78 & 21 & 0 \\ 3 & 4 & -77 & 356 & -77 & 4 & 3 \\ 0 & 21 & -78 & -77 & 8 & -11 & 3 \\ 0 & 0 & 21 & 4 & -11 & 6 & 0 \\ 0 & 0 & 0 & 3 & 3 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-2
Dual Refinable Mask Info	4-Compatible In Rhomb-2
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q1_NI_12_R2_4C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

69 Wavelet system: Q1_NI_16_C2_2C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 6$, $sm_2(a) = 6.0$

$$a : \quad \frac{1}{64} \begin{pmatrix} 1 & 3 & 3 & 1 \\ 3 & \boxed{9} & 9 & 3 \\ 3 & 9 & 9 & 3 \\ 1 & 3 & 3 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{128} \begin{pmatrix} -1 & -1 & -3 & -3 & 0 \\ -1 & -10 & -12 & -6 & -3 \\ -3 & -12 & \boxed{110} & -12 & -3 \\ -3 & -6 & -12 & -10 & -1 \\ 0 & -3 & -3 & -1 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{128} \begin{pmatrix} 0 & -3 & -3 & -1 & -1 \\ -3 & -6 & \boxed{-12} & -10 & -1 \\ -3 & -12 & 110 & -12 & -3 \\ -1 & -10 & -12 & -6 & -3 \\ -1 & -1 & -3 & -3 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q1_NI_16_C2_2C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

70 Wavelet system: Q1_NI_16_C2_4C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 6$, $sm_2(a) = 6.0$

$$a : \quad \frac{1}{64} \begin{pmatrix} 1 & 3 & 3 & 1 \\ 3 & \boxed{9} & 9 & 3 \\ 3 & 9 & 9 & 3 \\ 1 & 3 & 3 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4$, $sm_2(\tilde{a}) = 1.3524$

$$\tilde{a} : \quad \frac{1}{64} \begin{pmatrix} -5 & -3 & -3 & -5 \\ -3 & \boxed{27} & 27 & -3 \\ -3 & 27 & 27 & -3 \\ -5 & -3 & -3 & -5 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{2048} \begin{pmatrix} 5 & 3 & 18 & 14 & 9 & 15 & 0 \\ 3 & 18 & 9 & -36 & -27 & 18 & 15 \\ 18 & 9 & -207 & -270 & -108 & -27 & 9 \\ 14 & -36 & -270 & \boxed{1608} & -270 & -36 & 14 \\ 9 & -27 & -108 & -270 & -207 & 9 & 18 \\ 15 & 18 & -27 & -36 & 9 & 18 & 3 \\ 0 & 15 & 9 & 14 & 18 & 3 & 5 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{2048} \begin{pmatrix} 0 & 15 & 9 & 14 & 18 & 3 & 5 \\ 15 & 18 & -27 & -36 & 9 & 18 & 3 \\ 9 & -27 & -108 & \boxed{-270} & -207 & 9 & 18 \\ 14 & -36 & -270 & 1608 & -270 & -36 & 14 \\ 18 & 9 & -207 & -270 & -108 & -27 & 9 \\ 3 & 18 & 9 & -36 & -27 & 18 & 15 \\ 5 & 3 & 18 & 14 & 9 & 15 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-2
Dual Refinable Mask Info	4-Compatible In Cube-2
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q1_NI_16_C2_4C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

71 Wavelet system: Q1_NI_08_SS_4C_D4SXY_TF

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.03654$

$$a : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4$, $sm_2(\tilde{a}) = 3.03654$

$$\tilde{a} : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 4$, $VM(b_2) = 2$, $VM(b_3) = 2$,

$$b_1 : \quad \begin{pmatrix} -0.031 & 0 & 0 & 0.031 \\ 0 & 0.281 & \boxed{-0.281} & 0 \\ 0 & -0.281 & 0.281 & 0 \\ 0.031 & 0 & 0 & -0.031 \end{pmatrix}$$

$$b_2 : \quad \begin{pmatrix} -0.008 & 0 & 0 & -0.008 \\ 0 & \boxed{0.133} & 0.133 & 0 \\ 0 & -0.242 & -0.242 & 0 \\ 0.117 & 0 & 0 & 0.117 \end{pmatrix}$$

$$b_3 : \quad \begin{pmatrix} 0.117 & 0 & 0 & -0.117 \\ 0 & -0.242 & \boxed{0.242} & 0 \\ 0 & -0.133 & 0.133 & 0 \\ 0.008 & 0 & 0 & -0.008 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 2$, $VM(\tilde{b}_3) = 2$,

$$\tilde{b}_1 : \quad \begin{pmatrix} -0.031 & 0 & 0 & 0.031 \\ 0 & 0.281 & \boxed{-0.281} & 0 \\ 0 & -0.281 & 0.281 & 0 \\ 0.031 & 0 & 0 & -0.031 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \begin{pmatrix} -0.008 & 0 & 0 & -0.008 \\ 0 & \boxed{0.133} & 0.133 & 0 \\ 0 & -0.242 & -0.242 & 0 \\ 0.117 & 0 & 0 & 0.117 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \begin{pmatrix} 0.117 & 0 & 0 & -0.117 \\ 0 & -0.242 & \boxed{0.242} & 0 \\ 0 & -0.133 & 0.133 & 0 \\ 0.008 & 0 & 0 & -0.008 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Tight Wavelet Frame
Refinable Mask Info	Custom In Cube-2
Dual Refinable Mask Info	4-Compatible In Cube-2
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q1_NI_08_SS_4C_D4SXY_TF

Source: Created by the algorithm in paper "Symmetric Canonical Quincunx Tight Framelets with High Vanishing Moments and Smoothness" by B. Han, Q. Jiang, Z. Shen, X. Zhuang, 2015, Mathematics of Computation, 87 (309), 347-379

72 Wavelet system: Q1_NI_08_SS_2C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.03654$

$$a : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{64} \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & -8 & -9 & 0 & 0 \\ 0 & -9 & \boxed{46} & -9 & 0 \\ 0 & 0 & -9 & -8 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & \boxed{-9} & -8 & 1 \\ 0 & -9 & 46 & -9 & 0 \\ 1 & -8 & -9 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom In Cube-2
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q1_NI_08_SS_2C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

73 Wavelet system: Q1_NI_08_SS_4C_D4SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.03654$

$$a : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4$, $sm_2(\tilde{a}) = 3.03654$

$$\tilde{a} : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} -1 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 18 & 9 & 0 & 9 & 0 & 0 \\ 0 & 9 & -63 & -81 & 0 & 9 & 0 \\ -1 & 0 & -81 & \boxed{348} & -81 & 0 & -1 \\ 0 & 9 & 0 & -81 & -63 & 9 & 0 \\ 0 & 0 & 9 & 0 & 9 & 18 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & -1 \\ 0 & 0 & 9 & 0 & 9 & 18 & 0 \\ 0 & 9 & 0 & \boxed{-81} & -63 & 9 & 0 \\ -1 & 0 & -81 & 348 & -81 & 0 & -1 \\ 0 & 9 & -63 & -81 & 0 & 9 & 0 \\ 0 & 18 & 9 & 0 & 9 & 0 & 0 \\ -1 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom In Cube-2
Dual Refinable Mask Info	4-Compatible In Cube-2
Symmetry Info	D4 With Center (1/2,1/2)
Short Code	Q1_NI_08_SS_4C_D4SXY_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

74 Wavelet system: Q1_NI_09_SS_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.42789$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 1 & 4 & \boxed{12} & 4 & 1 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{128} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -4 & -1 & 0 & 0 \\ 0 & -1 & 0 & -13 & 0 & -1 & 0 \\ -1 & -4 & -13 & \boxed{80} & -13 & -4 & -1 \\ 0 & -1 & 0 & -13 & 0 & -1 & 0 \\ 0 & 0 & -1 & -4 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -2 & \boxed{-4} & -2 & 0 \\ -1 & -4 & 28 & -4 & -1 \\ 0 & -2 & -4 & -2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Special
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q1_NI_09_SS_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

75 Wavelet system: Q1_NI_09_SS_4C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.42789$

$$a : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 1 & 4 & \boxed{12} & 4 & 1 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.686754$

$$\tilde{a} : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & -9 & 0 & 0 \\ 0 & 6 & 8 & 6 & 0 \\ -9 & 8 & \boxed{44} & 8 & -9 \\ 0 & 6 & 8 & 6 & 0 \\ 0 & 0 & -9 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -6 & -8 & -6 & 0 & 0 & 0 \\ 0 & 0 & 18 & -8 & 64 & -8 & 18 & 0 & 0 \\ 0 & -6 & -8 & -84 & -104 & -84 & -8 & -6 & 0 \\ 9 & -8 & 64 & -104 & \boxed{532} & -104 & 64 & -8 & 9 \\ 0 & -6 & -8 & -84 & -104 & -84 & -8 & -6 & 0 \\ 0 & 0 & 18 & -8 & 64 & -8 & 18 & 0 & 0 \\ 0 & 0 & 0 & -6 & -8 & -6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 3 & -8 & 3 & 0 & 0 \\ 0 & 3 & -16 & \boxed{-47} & -16 & 3 & 0 \\ 9 & -8 & -47 & 224 & -47 & -8 & 9 \\ 0 & 3 & -16 & -47 & -16 & 3 & 0 \\ 0 & 0 & 3 & -8 & 3 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Special
Dual Refinable Mask Info	4-Compatible In Rhomb-2
Symmetry Info	D4
Short Code	Q1_NI_09_SS_4C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

76 Wavelet system: Q1_NI_13_R2S_4C_D4_FR

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.91803$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 8 & 2 & 0 \\ 1 & 8 & \boxed{20} & 8 & 1 \\ 0 & 2 & 8 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57796$

$$\tilde{a} : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 0 & -9 & 0 & 0 & 0 \\ 0 & 0 & -27 & 12 & -27 & 0 & 0 \\ 0 & -27 & 24 & 127 & 24 & -27 & 0 \\ -9 & 12 & 127 & \boxed{112} & 127 & 12 & -9 \\ 0 & -27 & 24 & 127 & 24 & -27 & 0 \\ 0 & 0 & -27 & 12 & -27 & 0 & 0 \\ 0 & 0 & 0 & -9 & 0 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$, $VM(b_2) = 2$,

$$b_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & -5 & 0 & 0 \\ 0 & -10 & -40 & -10 & 0 \\ -5 & -40 & \boxed{220} & -40 & -5 \\ 0 & -10 & -40 & -10 & 0 \\ 0 & 0 & -5 & 0 & 0 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -3 & -8 & -3 & 0 & 0 \\ 0 & -3 & -16 & \boxed{7} & -16 & -3 & 0 \\ -1 & -8 & 7 & 96 & 7 & -8 & -1 \\ 0 & -3 & -16 & 7 & -16 & -3 & 0 \\ 0 & 0 & -3 & -8 & -3 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{320} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 9 & -4 & 9 & 0 & 0 \\ 0 & 9 & -8 & -85 & -8 & 9 & 0 \\ 3 & -4 & -85 & \boxed{304} & -85 & -4 & 3 \\ 0 & 9 & -8 & -85 & -8 & 9 & 0 \\ 0 & 0 & 9 & -4 & 9 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} 0 & 0 & -3 & 0 & 0 \\ 0 & -6 & \boxed{4} & -6 & 0 \\ -3 & 4 & 20 & 4 & -3 \\ 0 & -6 & 4 & -6 & 0 \\ 0 & 0 & -3 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Custom in Rhomb-2
Dual Refinable Mask Info	4-Compatible In Rhomb-3
Symmetry Info	D4
Short Code	Q1_NI_13_R2S_4C_D4_FR

Source: Created by the algorithm in paper "Bi-frames with 4-fold axial symmetry for quadrilateral surface multiresolution processing" by Q. Jiang, 2010, CAM 234 (12), 3303-3325

77 Wavelet system: Q1_NI_13_R2S_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.91803$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 8 & 2 & 0 \\ 1 & 8 & \boxed{20} & 8 & 1 \\ 0 & 2 & 8 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 4$, $sm_2(\tilde{a}) = 4.0$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & \boxed{4} & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & -1 & -2 & -1 & 0 & 0 \\ 0 & -2 & -6 & -8 & -6 & -2 & 0 \\ -1 & -6 & -30 & -50 & -30 & -6 & -1 \\ -2 & -8 & -50 & \boxed{424} & -50 & -8 & -2 \\ -1 & -6 & -30 & -50 & -30 & -6 & -1 \\ 0 & -2 & -6 & -8 & -6 & -2 & 0 \\ 0 & 0 & -1 & -2 & -1 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & -1 & -2 & -1 & 0 \\ -1 & -4 & \boxed{-6} & -4 & -1 \\ -2 & -6 & 56 & -6 & -2 \\ -1 & -4 & -6 & -4 & -1 \\ 0 & -1 & -2 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom in Rhomb-2
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D4
Short Code	Q1_NI_13_R2S_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

78 Wavelet system: Q1_NI_13_R2S_4C_D4_FL

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 3.91803$

$$a : \quad \frac{1}{64} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 8 & 2 & 0 \\ 1 & 8 & \boxed{20} & 8 & 1 \\ 0 & 2 & 8 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = -1.45625$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} -3 & 2 & -3 \\ 2 & \boxed{20} & 2 \\ -3 & 2 & -3 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & 0 & 3 & -2 & 3 & 0 & 0 \\ 0 & 6 & -6 & -8 & -6 & 6 & 0 \\ 3 & -6 & 26 & -50 & 26 & -6 & 3 \\ -2 & -8 & -50 & \boxed{136} & -50 & -8 & -2 \\ 3 & -6 & 26 & -50 & 26 & -6 & 3 \\ 0 & 6 & -6 & -8 & -6 & 6 & 0 \\ 0 & 0 & 3 & -2 & 3 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 0 & 3 & -2 & 3 & 0 \\ 3 & -4 & \boxed{-14} & -4 & 3 \\ -2 & -14 & 56 & -14 & -2 \\ 3 & -4 & -14 & -4 & 3 \\ 0 & 3 & -2 & 3 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom in Rhomb-2
Dual Refinable Mask Info	4-Compatible In Cube-1
Symmetry Info	D4
Short Code	Q1_NI13_R2S_4C_D4_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

79 Wavelet system: SD_IN_02_ST_DF_NS_FL

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 1, sm_2(a) = 1.0$

$a : \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Dual Refinable Mask: $SR(\tilde{a}) = 0, sm_2(\tilde{a}) = -inf$

$\tilde{a} : \begin{pmatrix} 1 \end{pmatrix}$

Wavelet Masks: $VM(b_1) = 0,$

$b_1 : \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1,$

$\tilde{b}_1 : \frac{1}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Haar
Dual Refinable Mask Info	Delta Function
Symmetry Info	No Symmetry
Short Code	SD_IN_02_ST_DF_NS_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

80 Wavelet system: SD_IN_02_ST_BI_D2SX_OB

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 1.0$

$$a : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	ONB
Refinable Mask Info	Haar
Dual Refinable Mask Info	Haar
Symmetry Info	D2 With Center (1/2,0)
Short Code	SD_IN_02_ST_BI_D2SX_OB

Source: Haar wavelet

81 Wavelet system: SD_IN_03_HL_DF_D2_FL

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0, sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0,$

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	Delta Function
Symmetry Info	D2
Short Code	SD_IN_03_HL_DF_D2_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

82 Wavelet system: SD_IN_03_HL_1C_NS_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1, VM(\tilde{b}_2) = 1,$

$$\tilde{b}_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 3 \\ -1 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	1-Compatible In 2-point Standard
Symmetry Info	No Symmetry
Short Code	SD_IN_03_HL_1C_NS_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

83 Wavelet system: SD_IN_03_HL_2C_D2_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 \\ \boxed{2} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} -1 \\ \boxed{-2} \\ 6 \\ -2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	2-Compatible In 3-point Line
Symmetry Info	D2
Short Code	SD_IN_03_HL_2C_D2_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

84 Wavelet system: SD_IN_03_HL_2C_D2_FLL2_2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{0} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{4} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{0} & -1 \\ 0 & 6 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D2
Short Code	SD_IN_03_HL_2C_D2_FLL2_2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

85 Wavelet system: SD_IN_03_HL_BI_D2_FR

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.440765$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} -1 \\ 2 \\ \boxed{6} \\ 2 \\ -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$,

$$b_1 : \quad \frac{1}{8} \begin{pmatrix} -1 \\ \boxed{-2} \\ 6 \\ -2 \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In 3-point Line
Dual Refinable Mask Info	Biorthogonal In Rhomb-2
Symmetry Info	D2
Short Code	SD_IN_03_HL_BI_D2_FR

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

86 Wavelet system: SD_IN_03_C1_DF_D1_FL

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 0 & 0 & 1 \\ 0 & \boxed{2} & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0, sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0,$

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} 0 & \boxed{0} & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	Delta Function
Symmetry Info	D1
Short Code	SD_IN_03_C1_DF_D1_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

87 Wavelet system: SD_IN_03_C1_2C_D1_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 0 & 0 & 1 \\ 0 & \boxed{2} & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 \\ \boxed{2} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} 0 & 0 & -1 \\ 0 & \boxed{0} & -2 \\ -1 & 8 & -1 \\ -2 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D1
Short Code	SD_IN_03_C1_2C_D1_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

88 Wavelet system: SD_IN_03_C1_2C_D1_FLL2_2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 0 & 0 & 1 \\ 0 & \boxed{2} & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{0} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{4} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & \boxed{-1} & 0 & -1 \\ 0 & -1 & 8 & -1 & 0 \\ -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D1
Short Code	SD_IN_03_C1_2C_D1_FLL2_2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

89 Wavelet system: SD_IN_03_C1_BI_D1_FS

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 0 & 0 & 1 \\ 0 & \boxed{2} & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = -1.22299$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ -1 & 0 & \boxed{8} & 0 & -1 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & \boxed{0} & 0 & 1 \\ 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{16} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 2 & \boxed{0} & -8 & 0 & 1 \\ 0 & -2 & 0 & 16 & 0 & -2 & 0 \\ 1 & 0 & -8 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Dual Wavelet Frame In Sobolev Spaces
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	Biorthogonal In Rhomb-2
Symmetry Info	D1
Short Code	SD_IN_03_C1_BI_D1_FS

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

90 Wavelet system: SD_IN_03_C1_BLD1_FR

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 0 & 0 & 1 \\ 0 & \boxed{2} & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 0.440765$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & \boxed{6} & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & \boxed{2} & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{16} \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & \boxed{0} & -5 & 0 & 0 \\ 0 & 0 & 0 & 12 & 0 & 0 & 0 \\ 0 & 0 & -5 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	Biorthogonal In Cube-2
Symmetry Info	D1
Short Code	SD_IN_03_C1_BLD1_FR

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

91 Wavelet system: SD_IN_13_R3_DF_D2_FL

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.47603$

$$a : \quad \frac{1}{192} \begin{pmatrix} 0 & 0 & -6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & 48 & 4 & -1 \\ 0 & 0 & \boxed{96} & 0 & 0 \\ -1 & 4 & 48 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -6 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{192} \begin{pmatrix} 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & -4 & \boxed{-48} & -4 & 1 \\ 0 & 0 & 96 & 0 & 0 \\ 1 & -4 & -48 & -4 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	Delta Function
Symmetry Info	D2
Short Code	SD_IN_13_R3_DF_D2_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

92 Wavelet system: SD_IN_13_R3_2C_D2_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.47603$

$$a : \quad \frac{1}{192} \begin{pmatrix} 0 & 0 & -6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & 48 & 4 & -1 \\ 0 & 0 & \boxed{96} & 0 & 0 \\ -1 & 4 & 48 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -6 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 \\ \boxed{2} \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{384} \begin{pmatrix} 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 12 & 0 & 0 \\ 1 & -4 & -42 & -4 & 1 \\ 2 & -8 & \boxed{-96} & -8 & 2 \\ 2 & -8 & 288 & -8 & 2 \\ 2 & -8 & -96 & -8 & 2 \\ 1 & -4 & -42 & -4 & 1 \\ 0 & 0 & 12 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	2-Compatible In Rhomb-3
Symmetry Info	D2
Short Code	SD_IN_13_R3_2C_D2_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

93 Wavelet system: SD_NI_05_R2_2C_D2_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 4, sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 \\ 4 \\ \boxed{6} \\ 4 \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2, sm_2(\tilde{a}) = 2.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 \\ \boxed{2} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0,$

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} -1 \\ -2 \\ -7 \\ \boxed{20} \\ -7 \\ -2 \\ -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} -1 \\ \boxed{-2} \\ 6 \\ -2 \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-2
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D2
Short Code	SD_NI_05_R2_2C_D2_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

94 Wavelet system: SD_NI_05_R2_BI_D2_FS

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 4.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 \\ 4 \\ \boxed{6} \\ 4 \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = -1.64974$

$$\tilde{a} : \quad \frac{1}{32} \begin{pmatrix} 3 \\ -12 \\ 5 \\ \boxed{40} \\ 5 \\ -12 \\ 3 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{256} \begin{pmatrix} -3 \\ 12 \\ -23 \\ 32 \\ -38 \\ \boxed{40} \\ -38 \\ 32 \\ -23 \\ 12 \\ -3 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} -3 \\ 12 \\ -8 \\ \boxed{-28} \\ 54 \\ -28 \\ -8 \\ 12 \\ -3 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Dual Wavelet Frame In Sobolev Spaces
Refinable Mask Info	Best In Rhomb-2
Dual Refinable Mask Info	Biorthogonal In Rhomb-3
Symmetry Info	D2
Short Code	SD_NL05_R2_BI_D2_FS

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

95 Wavelet system: SD_NI_04_C1_2C_D1SX_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 1.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 & \boxed{1} & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.5$

$$\tilde{a} : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 1$,

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 & \boxed{2} & -1 \\ -1 & 2 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} 1 & \boxed{4} & -1 \\ 1 & -4 & -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D1 With Center (1/2,0)
Short Code	SD_NI_04_C1_2C_D1SX_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

96 Wavelet system: SD_NI_04_C1_2C_D2SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 1, sm_2(a) = 1.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 1,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 & \boxed{2} & -1 \\ -1 & 2 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D2 With Center (1/2,1/2)
Short Code	SD_NI_04_C1_2C_D2SXY_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

97 Wavelet system: SD_NI_04_C1_2C_D2SXY_TF

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 1, sm_2(a) = 1.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2, VM(b_2) = 1, VM(b_3) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} 1 & \boxed{-1} \\ -1 & 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ -1 & -1 \end{pmatrix}$$

$$b_3 : \quad \frac{1}{4} \begin{pmatrix} -1 & \boxed{1} \\ -1 & 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 1, VM(\tilde{b}_3) = 1,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} 1 & \boxed{-1} \\ -1 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ -1 & -1 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{4} \begin{pmatrix} -1 & \boxed{1} \\ -1 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Tight Wavelet Frame
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D2 With Center (1/2,1/2)
Short Code	SD_NI_04_C1_2C_D2SXY_TF

Source: Created by the algorithm in paper "Symmetric Canonical Quincunx Tight Framelets with High Vanishing Moments and Smoothness" by B. Han, Q. Jiang, Z. Shen, X. Zhuang, 2015, Mathematics of Computation, 87 (309), 347-379

98 Wavelet system: SD_NI_08_R2_2C_D2SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 3, sm_2(a) = 3.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 1 & 1 \\ \boxed{3} & 3 \\ 3 & 3 \\ 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 3,$

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} -1 & -2 & -1 \\ -1 & -2 & -1 \\ -6 & \boxed{20} & -6 \\ -6 & 20 & -6 \\ -1 & -2 & -1 \\ -1 & -2 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} -1 & -2 & -1 \\ -1 & -2 & -1 \\ 0 & \boxed{32} & 0 \\ 0 & -32 & 0 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-2
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D2 With Center (1/2,1/2)
Short Code	SD_NI_08_R2_2C_D2SXY_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

99 Wavelet system: SD_NI_12_R2_2C_D2SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 3, sm_2(a) = 3.0$

$$a : \quad \frac{1}{16} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & \boxed{2} & 2 & 1 \\ 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 3,$

$$\tilde{b}_1 : \quad \frac{1}{32} \begin{pmatrix} 0 & -1 & -2 & -1 & 0 \\ 0 & -1 & -2 & -1 & 0 \\ -2 & -6 & \boxed{24} & -6 & -2 \\ -2 & -6 & 24 & -6 & -2 \\ 0 & -1 & -2 & -1 & 0 \\ 0 & -1 & -2 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{32} \begin{pmatrix} -1 & -2 & -1 \\ -1 & -2 & -1 \\ 0 & \boxed{32} & 0 \\ 0 & -32 & 0 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-2
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D2 With Center (1/2,1/2)
Short Code	SD_NI_12_R2_2C_D2SXY_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

100 Wavelet system: SD_NI_08_SS_2C_D2SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.795618$

$$a : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 1$,

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{64} \begin{pmatrix} 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & -18 & \boxed{28} & -18 & 0 \\ 0 & -18 & 28 & -18 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{64} \begin{pmatrix} 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & \boxed{64} & 0 & 0 \\ 0 & 0 & -64 & 0 & 0 \\ -1 & -1 & 0 & -1 & -1 \\ -1 & -1 & 0 & -1 & -1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom In Cube-2
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	D2 With Center (1/2,1/2)
Short Code	SD_NI.08_SS_2C_D2SXY_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

101 Wavelet system: SD_NI_08_SS_4C_D2SXY_FLL2

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.795618$

$$a : \quad \frac{1}{32} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & \boxed{9} & 9 & 0 \\ 0 & 9 & 9 & 0 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.795618$

$$\tilde{a} : \quad \frac{1}{32} \begin{pmatrix} 0 & -1 & -1 & 0 \\ -1 & \boxed{10} & 10 & -1 \\ -1 & 10 & 10 & -1 \\ 0 & -1 & -1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 1$,

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ 1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{512} \begin{pmatrix} 0 & -1 & -1 & 0 & -1 & -1 & 0 \\ -1 & 10 & 10 & -2 & 10 & 10 & -1 \\ -1 & 10 & 10 & 34 & 28 & 10 & -1 \\ 0 & 17 & -163 & \boxed{152} & -163 & 17 & 0 \\ 0 & 17 & -163 & 152 & -163 & 17 & 0 \\ -1 & 10 & 28 & 34 & 28 & 10 & -1 \\ -1 & 10 & 10 & -2 & 10 & 10 & -1 \\ 0 & -1 & -1 & 0 & -1 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{512} \begin{pmatrix} 0 & -1 & -1 & 0 & -1 & -1 & 0 \\ -1 & 10 & 10 & -2 & 10 & 10 & -1 \\ -1 & 10 & 10 & -2 & 10 & 10 & -1 \\ 0 & -1 & -1 & \boxed{512} & -1 & -1 & 0 \\ 0 & 1 & 1 & -512 & 1 & 1 & 0 \\ 1 & -10 & -10 & 2 & -10 & -10 & 1 \\ 1 & -10 & -10 & 2 & -10 & -10 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Custom In Cube-2
Dual Refinable Mask Info	4-Compatible In Cube-2
Symmetry Info	D2 With Center (1/2,1/2)
Short Code	SD_NL08_SS_4C_D2SXY_FLL2

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

102 Wavelet system: H0_IN_03_HR_DF_NS_FL

Dilation Matrix: $\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.36907$

$a : \quad \frac{1}{3} \begin{pmatrix} 1 & \boxed{1} \\ 0 & 1 \end{pmatrix}$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$

$b_2 : \quad (1)$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$,

$\tilde{b}_1 : \quad \frac{1}{3} \begin{pmatrix} \boxed{-1} \\ 1 \end{pmatrix}$

$\tilde{b}_2 : \quad \frac{1}{3} \begin{pmatrix} 1 & \boxed{-1} \end{pmatrix}$

Additional Information:

Property	Value
Dilation Matrix	HexagonalRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Haar
Dual Refinable Mask Info	Delta Function
Symmetry Info	Only Refinable
Short Code	H0_IN_03_HR_DF_NS_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

103 Wavelet system: H0_IN_03_HR_BI_HA_FR

Dilation Matrix: $\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.36907$

$$a : \quad \frac{1}{3} \begin{pmatrix} 1 & \boxed{1} \\ 0 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.36907$

$$\tilde{a} : \quad \frac{1}{3} \begin{pmatrix} 1 & \boxed{1} \\ 0 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$, $VM(b_2) = 1$,

$$b_1 : \quad \frac{1}{3} \begin{pmatrix} -1 & \boxed{-1} \\ 0 & 2 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{3} \begin{pmatrix} 2 & \boxed{-1} \\ 0 & -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{3} \begin{pmatrix} \boxed{-1} \\ 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{3} \begin{pmatrix} 1 & \boxed{-1} \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	HexagonalRotation
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Haar
Dual Refinable Mask Info	Biorthogonal In The Same Support
Symmetry Info	HexAbel
Short Code	H0_IN_03_HR_BI_HA_FR

Source: Created by the algorithm in paper "Symmetric multivariate wavelets" by Karakazyan S., Skopina M., Tchobanou M. in Int. J. Wavelets Multiresolut. Inf. Process. 2009 7 (3) P. 313-340.

104 Wavelet system: H0_IN_03_HR_BI_HA_OB

Dilation Matrix: $\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.36907$

$$a : \quad \frac{1}{3} \begin{pmatrix} 1 & \boxed{1} \\ 0 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.36907$

$$\tilde{a} : \quad \frac{1}{3} \begin{pmatrix} 1 & \boxed{1} \\ 0 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$, $VM(b_2) = 1$,

$$b_1 : \quad \begin{pmatrix} 0.122 & \boxed{0.333} \\ 0 & -0.455 \end{pmatrix}$$

$$b_2 : \quad \begin{pmatrix} -0.455 & \boxed{0.333} \\ 0 & 0.122 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \begin{pmatrix} 0.122 & \boxed{0.333} \\ 0 & -0.455 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \begin{pmatrix} -0.455 & \boxed{0.333} \\ 0 & 0.122 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	HexagonalRotation
Wavelet System Type	ONB
Refinable Mask Info	Haar
Dual Refinable Mask Info	Biorthogonal In The Same Support
Symmetry Info	HexAbel
Short Code	H0_IN_03_HR_BI_HA_OB

Source: Thesis

105 Wavelet system: H0_IN_03_HL_DF_NS_FL

Dilation Matrix: $\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.17122$

$$a : \quad \frac{1}{3} \begin{pmatrix} 1 \\ \boxed{1} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

$$b_2 : \quad (1)$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{3} \begin{pmatrix} \boxed{-1} \\ 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{3} \begin{pmatrix} 1 \\ \boxed{-1} \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	HexagonalRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Haar
Dual Refinable Mask Info	Delta Function
Symmetry Info	Only Refinable
Short Code	H0_IN_03_HL_DF_NS_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

106 Wavelet system: H0_IN_03_HL_BI_HA_FR

Dilation Matrix: $\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.17122$

$$a : \quad \frac{1}{3} \begin{pmatrix} 1 \\ \boxed{1} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.17122$

$$\tilde{a} : \quad \frac{1}{3} \begin{pmatrix} 1 \\ \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$, $VM(b_2) = 1$,

$$b_1 : \quad \frac{1}{3} \begin{pmatrix} -1 \\ \boxed{-1} \\ 2 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{3} \begin{pmatrix} 2 \\ \boxed{-1} \\ -1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{3} \begin{pmatrix} \boxed{-1} \\ 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{3} \begin{pmatrix} 1 \\ \boxed{-1} \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	HexagonalRotation
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Haar
Dual Refinable Mask Info	Biorthogonal In The Same Support
Symmetry Info	HexAbel
Short Code	H0_IN_03_HL_BI_HA_FR

Source: Created by the algorithm in paper "Symmetric multivariate wavelets" by Karakazyan S., Skopina M., Tchobanou M. in Int. J. Wavelets Multiresolut. Inf. Process. 2009 7 (3) P. 313-340.

107 Wavelet system: H0_IN_03_HL_BI_HA_OB

Dilation Matrix: $\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.17122$

$$a : \quad \frac{1}{3} \begin{pmatrix} 1 \\ \boxed{1} \\ 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.17122$

$$\tilde{a} : \quad \frac{1}{3} \begin{pmatrix} 1 \\ \boxed{1} \\ 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$, $VM(b_2) = 1$,

$$b_1 : \quad \begin{pmatrix} 0.122 \\ \boxed{0.333} \\ -0.455 \end{pmatrix}$$

$$b_2 : \quad \begin{pmatrix} -0.455 \\ \boxed{0.333} \\ 0.122 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$,

$$\tilde{b}_1 : \quad \begin{pmatrix} 0.122 \\ \boxed{0.333} \\ -0.455 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \begin{pmatrix} -0.455 \\ \boxed{0.333} \\ 0.122 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	HexagonalRotation
Wavelet System Type	ONB
Refinable Mask Info	Haar
Dual Refinable Mask Info	Biorthogonal In The Same Support
Symmetry Info	HexAbel
Short Code	H0_IN_03_HL_BI_HA_OB

Source: Thesis

108 Wavelet system: H1_IN_07_C1_DF_HA_FL

Dilation Matrix: $\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.65714$

$$a : \quad \frac{1}{9} \begin{pmatrix} 1 & 1 & 0 \\ 1 & \boxed{3} & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

$$b_2 : \quad (1)$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{9} \begin{pmatrix} 0 & \boxed{-1} & 0 \\ -1 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{9} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & \boxed{-1} & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	HexagonalRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	Delta Function
Symmetry Info	HexAbel
Short Code	H1_IN_07_C1_DF_HA_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

109 Wavelet system: H1_IN_07_C1_2C_HA_FR

Dilation Matrix: $\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 2$, $sm_2(a) = 1.65714$

$$a : \quad \frac{1}{9} \begin{pmatrix} 1 & 1 & 0 \\ 1 & \boxed{3} & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.65714$

$$\tilde{a} : \quad \frac{1}{9} \begin{pmatrix} 1 & 1 & 0 \\ 1 & \boxed{3} & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$, $VM(b_3) = 0$,

$$b_1 : \quad \frac{1}{3} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{3} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

$$b_3 : \quad \frac{1}{3} (1)$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$, $VM(\tilde{b}_3) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{9} \begin{pmatrix} -1 & -1 & 0 \\ -1 & \boxed{6} & -1 \\ 0 & -1 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{27} \begin{pmatrix} 0 & -1 & -1 & 0 & 0 \\ -1 & -2 & \boxed{-3} & -1 & 0 \\ -1 & -3 & 24 & -2 & 0 \\ 0 & -1 & -2 & -3 & -1 \\ 0 & 0 & 0 & -1 & -1 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{27} \begin{pmatrix} -1 & -1 & 0 & 0 & 0 \\ -1 & -3 & -2 & -1 & 0 \\ 0 & -2 & 24 & -3 & -1 \\ 0 & -1 & \boxed{-3} & -2 & -1 \\ 0 & 0 & -1 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	HexagonalRotation
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	2-Compatible In Cube-1
Symmetry Info	HexAbel
Short Code	H1_IN_07_C1_2C_HA_FR

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

110 Wavelet system: D0_IN_04_HR_DF_NS_FL

Dilation Matrix: $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

Refinable Mask: $SR(a) = 1, sm_2(a) = 1.0$

$a : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$

Dual Refinable Mask: $SR(\tilde{a}) = 0, sm_2(\tilde{a}) = -inf$

$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0, VM(b_3) = 0,$

$b_1 : \quad \begin{pmatrix} \boxed{0} & 1 \end{pmatrix}$

$b_2 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$

$b_3 : \quad \begin{pmatrix} \boxed{0} & 0 \\ 0 & 1 \end{pmatrix}$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1, VM(\tilde{b}_2) = 1, VM(\tilde{b}_3) = 1,$

$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} & 1 \end{pmatrix}$

$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 1 \end{pmatrix}$

$\tilde{b}_3 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} & 0 \\ 0 & 1 \end{pmatrix}$

Additional Information:

Property	Value
Dilation Matrix	DyadicStandard
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Haar
Dual Refinable Mask Info	Delta Function
Symmetry Info	Only Refinable
Short Code	D0_IN_04_HR_DF_NS_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

111 Wavelet system: D0_IN_04_HR_BI_D2_FR

Dilation Matrix: $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

Refinable Mask: $SR(a) = 1, sm_2(a) = 1.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1, VM(b_2) = 1, VM(b_3) = 1,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} & 3 \\ -1 & -1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} & -1 \\ 3 & -1 \end{pmatrix}$$

$$b_3 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} & -1 \\ -1 & 3 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1, VM(\tilde{b}_2) = 1, VM(\tilde{b}_3) = 1,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 1 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} & 0 \\ 0 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	DyadicStandard
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Haar
Dual Refinable Mask Info	Biorthogonal In The Same Support
Symmetry Info	D2 With Center (1/2,1/2)
Short Code	D0_IN_04_HR_BI_D2_FR

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

112 Wavelet system: D0_IN_04_HR_BI_D2_OB

Dilation Matrix: $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

Refinable Mask: $SR(a) = 1, sm_2(a) = 1.0$

$$a : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1, sm_2(\tilde{a}) = 1.0$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ 1 & 1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1, VM(b_2) = 1, VM(b_3) = 2,$

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ -1 & -1 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & -1 \\ 1 & -1 \end{pmatrix}$$

$$b_3 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & -1 \\ -1 & 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1, VM(\tilde{b}_2) = 1, VM(\tilde{b}_3) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & 1 \\ -1 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & -1 \\ 1 & -1 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{4} \begin{pmatrix} \boxed{1} & -1 \\ -1 & 1 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	DyadicStandard
Wavelet System Type	ONB
Refinable Mask Info	Haar
Dual Refinable Mask Info	Biorthogonal In The Same Support
Symmetry Info	D2 With Center (1/2,1/2)
Short Code	D0_IN_04_HR_BI_D2_OB

Source: Thesis

113 Wavelet system: D0_IN_07_C1_DF_D1_FL

Dilation Matrix: $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

Refinable Mask: $SR(a) = 2, sm_2(a) = 2.0$

$$a : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 1 \\ 1 & \boxed{2} & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0, sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0, VM(b_2) = 0, VM(b_3) = 0,$

$$b_1 : \quad \begin{pmatrix} \boxed{0} & 1 \end{pmatrix}$$

$$b_2 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

$$b_3 : \quad \begin{pmatrix} \boxed{0} & 0 \\ 0 & 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2, VM(\tilde{b}_2) = 2, VM(\tilde{b}_3) = 2,$

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} \boxed{-1} & 2 & -1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{8} \begin{pmatrix} \boxed{-1} \\ 2 \\ -1 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{8} \begin{pmatrix} \boxed{0} & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	DyadicStandard
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Cube-1
Dual Refinable Mask Info	Delta Function
Symmetry Info	D1
Short Code	D0_IN_07_C1_DF_D1_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

114 Wavelet system: D1_IN_04_HR_DF_NS_FL

Dilation Matrix: $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.207519$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \boxed{1} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \left(\boxed{1} \right)$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$, $VM(b_3) = 0$,

$$b_1 : \quad \left(\boxed{0} \quad 1 \right)$$

$$b_2 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

$$b_3 : \quad (1)$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$, $VM(\tilde{b}_3) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{4} \left(\boxed{-1} \quad 1 \right)$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 1 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{4} \begin{pmatrix} 1 & 0 \\ 0 & \boxed{-1} \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	DyadicNonStandard
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Haar
Dual Refinable Mask Info	Delta Function
Symmetry Info	No Symmetry
Short Code	D1_IN_04_HR_DF_NS_FL

Source: Created by the algorithm in paper "Approximation by frame-like wavelet systems" by A. Krivoshein, M. Skopina, 2011, ACHA 31 (3), 410-428

115 Wavelet system: D1_IN_04_HR_BI_NS_FR

Dilation Matrix: $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.207519$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \boxed{1} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.207519$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \boxed{1} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$, $VM(b_2) = 1$, $VM(b_3) = 1$,

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} -1 & 0 & 0 \\ 0 & \boxed{-1} & 3 \\ 0 & -1 & 0 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} -1 & 0 & 0 \\ 0 & \boxed{-1} & -1 \\ 0 & 3 & 0 \end{pmatrix}$$

$$b_3 : \quad \frac{1}{4} \begin{pmatrix} 3 & 0 & 0 \\ 0 & \boxed{-1} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$, $VM(\tilde{b}_3) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} \boxed{-1} \\ 1 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{4} \begin{pmatrix} 1 & 0 \\ 0 & \boxed{-1} \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	DyadicNonStandard
Wavelet System Type	Dual Wavelet Frame
Refinable Mask Info	Haar
Dual Refinable Mask Info	Biorthogonal In The Same Support
Symmetry Info	No Symmetry
Short Code	D1_IN_04_HR_BI_NS_FR

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

116 Wavelet system: D1_IN_04_HR_BI_NS_OB

Dilation Matrix: $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

Refinable Mask: $SR(a) = 1$, $sm_2(a) = 0.207519$

$$a : \quad \frac{1}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \boxed{1} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 1$, $sm_2(\tilde{a}) = 0.207519$

$$\tilde{a} : \quad \frac{1}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \boxed{1} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 1$, $VM(b_2) = 1$, $VM(b_3) = 1$,

$$b_1 : \quad \frac{1}{4} \begin{pmatrix} -1 & 0 & 0 \\ 0 & \boxed{1} & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$b_2 : \quad \frac{1}{4} \begin{pmatrix} -1 & 0 & 0 \\ 0 & \boxed{1} & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$b_3 : \quad \frac{1}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \boxed{1} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 1$, $VM(\tilde{b}_2) = 1$, $VM(\tilde{b}_3) = 1$,

$$\tilde{b}_1 : \quad \frac{1}{4} \begin{pmatrix} -1 & 0 & 0 \\ 0 & \boxed{1} & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{4} \begin{pmatrix} -1 & 0 & 0 \\ 0 & \boxed{1} & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\tilde{b}_3 : \quad \frac{1}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \boxed{1} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	DyadicNonStandard
Wavelet System Type	ONB
Refinable Mask Info	Haar
Dual Refinable Mask Info	Biorthogonal In The Same Support
Symmetry Info	No Symmetry
Short Code	D1_IN_04_HR_BI_NS_OB

Source: Thesis

117 Wavelet system: Q0_IN_17_R3_DF_D4_FL

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.62673$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 3 & 0 & 43 & \boxed{128} & 43 & 0 & 3 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 7 & 0 & 0 \\ 0 & 7 & 0 & \boxed{-43} & 0 & 7 & 0 \\ -3 & 0 & -43 & 128 & -43 & 0 & -3 \\ 0 & 7 & 0 & -43 & 0 & 7 & 0 \\ 0 & 0 & 7 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	Delta Function
Symmetry Info	D4
Short Code	Q0_IN_17_R3_DF_D4_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

118 Wavelet system: Q0_IN_17_R3_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.62673$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 3 & 0 & 43 & \boxed{128} & 43 & 0 & 3 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{4} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & -12 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 14 & 28 & -32 & 28 & 14 & 0 & 0 & 0 \\ 0 & 4 & 28 & -72 & \boxed{-172} & -72 & 28 & 4 & 0 & 0 \\ -3 & -12 & -32 & -172 & 852 & -172 & -32 & -12 & -3 & -3 \\ 0 & 4 & 28 & -72 & -172 & -72 & 28 & 4 & 0 & 0 \\ 0 & 0 & 14 & 28 & -32 & 28 & 14 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & -12 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q0_IN_17_R3_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

119 Wavelet system: Q0_IN_17_R3_4C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.62673$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 3 & 0 & 43 & \boxed{128} & 43 & 0 & 3 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.235724$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} -1 & 2 & -1 \\ 2 & \boxed{12} & 2 \\ -1 & 2 & -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{16} \begin{pmatrix} 1 & -2 & 1 \\ -2 & \boxed{4} & -2 \\ 1 & -2 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{2048} \begin{pmatrix} 0 & 0 & 0 & 3 & -6 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 8 & -50 & 8 & -7 & 0 & 0 \\ 0 & -7 & 28 & 123 & -64 & 123 & 28 & -7 & 0 \\ 3 & 8 & 123 & -144 & \boxed{-444} & -144 & 123 & 8 & 3 \\ -6 & -50 & -64 & -444 & 1704 & -444 & -64 & -50 & -6 \\ 3 & 8 & 123 & -144 & -444 & -144 & 123 & 8 & 3 \\ 0 & -7 & 28 & 123 & -64 & 123 & 28 & -7 & 0 \\ 0 & 0 & -7 & 8 & -50 & 8 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & -6 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	4-Compatible In Cube-1
Symmetry Info	D4
Short Code	Q0_IN_17_R3_4C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

120 Wavelet system: Q0_IN_17_R3_BI_D4_FS

Dilation Matrix: $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.62673$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 3 & 0 & 43 & \boxed{128} & 43 & 0 & 3 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = -0.128504$

$$\tilde{a} : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 14 & 0 & -32 & 0 & 14 & 0 & 0 \\ 0 & 4 & 0 & -72 & 128 & -72 & 0 & 4 & 0 \\ -3 & 0 & -32 & 128 & \boxed{852} & 128 & -32 & 0 & -3 \\ 0 & 4 & 0 & -72 & 128 & -72 & 0 & 4 & 0 \\ 0 & 0 & 14 & 0 & -32 & 0 & 14 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$,

$$b_1 : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 14 & 0 & -32 & 0 & 14 & 0 & 0 \\ 0 & 4 & 0 & -72 & \boxed{-128} & -72 & 0 & 4 & 0 \\ -3 & 0 & -32 & -128 & 852 & -128 & -32 & 0 & -3 \\ 0 & 4 & 0 & -72 & -128 & -72 & 0 & 4 & 0 \\ 0 & 0 & 14 & 0 & -32 & 0 & 14 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 7 & 0 & 0 \\ 0 & 7 & 0 & \boxed{-43} & 0 & 7 & 0 \\ -3 & 0 & -43 & 128 & -43 & 0 & -3 \\ 0 & 7 & 0 & -43 & 0 & 7 & 0 \\ 0 & 0 & 7 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxRotation
Wavelet System Type	Dual Wavelet Frame In Sobolev Spaces
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	Biorthogonal In Rhomb-4
Symmetry Info	D4
Short Code	Q0_IN_17_R3_BI_D4_FS

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

121 Wavelet system: Q1_IN_17_R3_DF_D4_FL

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.62673$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 3 & 0 & 43 & \boxed{128} & 43 & 0 & 3 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 0$, $sm_2(\tilde{a}) = -inf$

$$\tilde{a} : \quad \begin{pmatrix} \boxed{1} \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$,

$$b_1 : \quad \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 7 & 0 & 0 \\ 0 & 7 & 0 & \boxed{-43} & 0 & 7 & 0 \\ -3 & 0 & -43 & 128 & -43 & 0 & -3 \\ 0 & 7 & 0 & -43 & 0 & 7 & 0 \\ 0 & 0 & 7 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	Delta Function
Symmetry Info	D4
Short Code	Q1_IN_17_R3_DF_D4_FL

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

122 Wavelet system: Q1_IN_17_R3_2C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.62673$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 3 & 0 & 43 & \boxed{128} & 43 & 0 & 3 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 1.57764$

$$\tilde{a} : \quad \frac{1}{8} \begin{pmatrix} 0 & 1 & 0 \\ 1 & \boxed{4} & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 2$, $VM(\tilde{b}_2) = 2$,

$$\tilde{b}_1 : \quad \frac{1}{8} \begin{pmatrix} 0 & -1 & 0 \\ -1 & \boxed{4} & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & -12 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 14 & 28 & -32 & 28 & 14 & 0 & 0 & 0 \\ 0 & 4 & 28 & -72 & \boxed{-172} & -72 & 28 & 4 & 0 & 0 \\ -3 & -12 & -32 & -172 & 852 & -172 & -32 & -12 & -3 & -3 \\ 0 & 4 & 28 & -72 & -172 & -72 & 28 & 4 & 0 & 0 \\ 0 & 0 & 14 & 28 & -32 & 28 & 14 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & -12 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	2-Compatible In Rhomb-1
Symmetry Info	D4
Short Code	Q1_IN_17_R3_2C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

123 Wavelet system: Q0_IN_17_R3_4C_D4_FLL2

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.62673$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 3 & 0 & 43 & \boxed{128} & 43 & 0 & 3 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.235724$

$$\tilde{a} : \quad \frac{1}{16} \begin{pmatrix} -1 & 2 & -1 \\ 2 & \boxed{12} & 2 \\ -1 & 2 & -1 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 0$, $VM(b_2) = 0$,

$$b_1 : \quad \frac{1}{2} \begin{pmatrix} \boxed{1} \end{pmatrix}$$

$$b_2 : \quad \frac{1}{2} \begin{pmatrix} \boxed{0} \\ 1 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$, $VM(\tilde{b}_2) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{16} \begin{pmatrix} 1 & -2 & 1 \\ -2 & \boxed{4} & -2 \\ 1 & -2 & 1 \end{pmatrix}$$

$$\tilde{b}_2 : \quad \frac{1}{2048} \begin{pmatrix} 0 & 0 & 0 & 3 & -6 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 8 & -50 & 8 & -7 & 0 & 0 \\ 0 & -7 & 28 & 123 & -64 & 123 & 28 & -7 & 0 \\ 3 & 8 & 123 & -144 & \boxed{-444} & -144 & 123 & 8 & 3 \\ -6 & -50 & -64 & -444 & 1704 & -444 & -64 & -50 & -6 \\ 3 & 8 & 123 & -144 & -444 & -144 & 123 & 8 & 3 \\ 0 & -7 & 28 & 123 & -64 & 123 & 28 & -7 & 0 \\ 0 & 0 & -7 & 8 & -50 & 8 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & -6 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	4-Compatible In Cube-1
Symmetry Info	D4
Short Code	Q0_IN_17_R3_4C_D4_FLL2

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

124 Wavelet system: Q1_IN_17_R3_BI_D4_FS

Dilation Matrix: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.62673$

$$a : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 3 & 0 & 43 & \boxed{128} & 43 & 0 & 3 \\ 0 & -7 & 0 & 43 & 0 & -7 & 0 \\ 0 & 0 & -7 & 0 & -7 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = -0.128504$

$$\tilde{a} : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 14 & 0 & -32 & 0 & 14 & 0 & 0 \\ 0 & 4 & 0 & -72 & 128 & -72 & 0 & 4 & 0 \\ -3 & 0 & -32 & 128 & \boxed{852} & 128 & -32 & 0 & -3 \\ 0 & 4 & 0 & -72 & 128 & -72 & 0 & 4 & 0 \\ 0 & 0 & 14 & 0 & -32 & 0 & 14 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$,

$$b_1 : \quad \frac{1}{1024} \begin{pmatrix} 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 14 & 0 & -32 & 0 & 14 & 0 & 0 \\ 0 & 4 & 0 & -72 & \boxed{-128} & -72 & 0 & 4 & 0 \\ -3 & 0 & -32 & -128 & 852 & -128 & -32 & 0 & -3 \\ 0 & 4 & 0 & -72 & -128 & -72 & 0 & 4 & 0 \\ 0 & 0 & 14 & 0 & -32 & 0 & 14 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{256} \begin{pmatrix} 0 & 0 & 0 & -3 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 7 & 0 & 0 \\ 0 & 7 & 0 & \boxed{-43} & 0 & 7 & 0 \\ -3 & 0 & -43 & 128 & -43 & 0 & -3 \\ 0 & 7 & 0 & -43 & 0 & 7 & 0 \\ 0 & 0 & 7 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	QuincunxReflection
Wavelet System Type	Dual Wavelet Frame In Sobolev Spaces
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	Biorthogonal In Rhomb-4
Symmetry Info	D4
Short Code	Q1_IN_17_R3_BI_D4_FS

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238

125 Wavelet system: SD_IN_13_R3_BI_D2_FR

Dilation Matrix: $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

Refinable Mask: $SR(a) = 4$, $sm_2(a) = 2.47603$

$$a : \quad \frac{1}{192} \begin{pmatrix} 0 & 0 & -6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & 48 & 4 & -1 \\ 0 & 0 & \boxed{96} & 0 & 0 \\ -1 & 4 & 48 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -6 & 0 & 0 \end{pmatrix}$$

Dual Refinable Mask: $SR(\tilde{a}) = 2$, $sm_2(\tilde{a}) = 0.444012$

$$\tilde{a} : \quad \frac{1}{384} \begin{pmatrix} 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & -4 & -42 & -4 & 1 \\ 0 & 0 & 96 & 0 & 0 \\ 2 & -8 & \boxed{288} & -8 & 2 \\ 0 & 0 & 96 & 0 & 0 \\ 1 & -4 & -42 & -4 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 \end{pmatrix}$$

Wavelet Masks: $VM(b_1) = 2$,

$$b_1 : \quad \frac{1}{384} \begin{pmatrix} 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & -4 & -42 & -4 & 1 \\ 0 & 0 & \boxed{-96} & 0 & 0 \\ 2 & -8 & 288 & -8 & 2 \\ 0 & 0 & -96 & 0 & 0 \\ 1 & -4 & -42 & -4 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 \end{pmatrix}$$

Dual Wavelet Masks: $VM(\tilde{b}_1) = 4$,

$$\tilde{b}_1 : \quad \frac{1}{192} \begin{pmatrix} 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & -4 & \boxed{-48} & -4 & 1 \\ 0 & 0 & 96 & 0 & 0 \\ 1 & -4 & -48 & -4 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 \end{pmatrix}$$

Additional Information:

Property	Value
Dilation Matrix	SemiDyadic
Wavelet System Type	Frame-like Dual Wavelet System
Refinable Mask Info	Best In Rhomb-3
Dual Refinable Mask Info	Biorthogonal In Rhomb-4
Symmetry Info	D2
Short Code	SD_IN_13_R3_BI_D2_FR

Source: Created by the algorithm in paper "On construction of multivariate symmetric MRA-based wavelets" by A. Krivoshein, 2014, ACHA 36 (2), 215-238